A symmetrized B_2 Bailey lemma and Rogers–Ramanujan-type identities

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Abstract. Motivated by the A_2 Bailey framework of Andrews, Schilling and Warnaar, we develop a Bailey method associated with the symmetrized B_2 quadratic form. The construction consists of a B_2 Bailey lemma, Bailey pairs, and compatible chain and lattice transformations. Its basic finite identity is

$$\sum_{r=0}^{N}\sum_{\ell=0}^{L}\frac{z^r w^\ell q^{r^2-2r\ell+2\ell^2}}{(q;q)_{N-r}(q^2;q^2)_{L-\ell}}\Psi_{r,\ell}(z,w;q)=\Psi_{N,L}(z,w;q).$$

We construct two initial Bailey pairs and use a normalized spinor character of $V_{B_2}(s\omega_{\mathrm{sp}})$, together with Jacobi and Macdonald identities, to derive families of Rogers–Ramanujan-type identities with A_1- and D_2-type theta evaluations. Applications include multisum expressions for principally specialized $A_1^{(1)}$ characters, extremal CMPP partition generating functions, and weighted sums of Hall–Littlewood polynomials.

Keywords: B_2 Bailey lemma, Rogers–Ramanujan-type identities, affine Lie algebra characters, Macdonald identities

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1 Introduction

We begin with the two classical Rogers–Ramanujan identities [35–37]:

$$1+\sum_{n=1}^{\infty}\frac{q^{n^2}}{(1-q)(1-q^2)\cdots(1-q^n)}=\prod_{n=1}^{\infty}\frac{1}{(1-q^{5n-1})(1-q^{5n-4})},$$
$$1+\sum_{n=1}^{\infty}\frac{q^{n(n+1)}}{(1-q)(1-q^2)\cdots(1-q^n)}=\prod_{n=1}^{\infty}\frac{1}{(1-q^{5n-2})(1-q^{5n-3})}.$$

From the perspective of partition theory, the Rogers–Ramanujan identities have nice interpretations. A partition λ of a nonnegative integer n is a finite nonincreasing sequence of positive integers $\lambda=(\lambda_1,\lambda_2,\ldots,\lambda_k)$ such that $|\lambda|=\lambda_1+\cdots+\lambda_k=n$. Let $i\in\{1,2\}$ and denote by $A_{2,i}(n)$ the number of partitions of n such that the difference between consecutive parts is at least 2 and the part 1 occurs at most $i-1$ times. Let $B_{2,i}(n)$ be the number of partitions of n into parts congruent to $\pm(2+i)$podmod 5. Then the Rogers–Ramanujan identities [31] assert that $A_{2,i}(n)=B_{2,i}(n)$.

Gordon later extended these identities to arbitrary odd moduli [16]. Andrews obtained the analytic odd-modulus extension of Gordon's identities [3], and Bressoud established the even-modulus counterpart [8]. Together these are the Andrews–Gordon–Bressoud (AGB) identities. For integers $k\geq 1$, $0\leq a\leq k$, and $\sigma\in\{0,1\}$, put $K:=2k+2+\sigma$. Then [2,5,39]

$$\sum_{\lambda_1\geq\cdots\geq\lambda_k\geq 0}\frac{q^{\lambda_1^2+\cdots+\lambda_k^2+\lambda_{a+1}+\cdots+\lambda_k}}{(q;q)_{\lambda_1-\lambda_2}\cdots(q;q)_{\lambda_{k-1}-\lambda_k}(q^{2-\sigma};q^{2-\sigma})_{\lambda_k}}=\frac{(q^{a+1},q^{K-a-1},q^K;q^K)_\infty}{(q;q)_\infty}. \tag{1.1}$$

Andrews interpreted the multisum in (1.1) using Durfee dissections [4]. Other partition-theoretic interpretations were given by Warnaar in the odd-modulus case [42] and by Dousse, Jouhet, and Konan for identities of Andrews–Gordon and Bressoud type [11]. Afsharijoo, Dousse, Jouhet, and Mourtada [1] showed that this multisum also arises naturally in a commutative-algebraic framework involving graded quotient rings, Durfee squares, and partition theory, thereby obtaining new companions to the classical AGB identities.

For fixed $\sigma \in \{0,1\}$, let $\chi_{a,\sigma}^{k-1}(q)$ denote the common value of the two sides of (1.1). These functions are closely related to specialized characters of affine Lie algebras. Let $\mathfrak{g}$ be an affine Lie algebra with simple roots indexed by $0,\dots,r$, and let $\mathfrak{h}$ and $\mathfrak{h}^*$ denote its Cartan subalgebra and its dual, respectively. Let $\alpha_0,\dots,\alpha_r$ be the simple roots, $\alpha_0^\vee,\dots,\alpha_r^\vee$ the simple coroots, and $\Lambda_0,\dots,\Lambda_r$ the fundamental weights. Let δ denote the null root. The fundamental weights satisfy $\langle \alpha_i^\vee, \Lambda_j\rangle = \delta_{ij}$; see [21].

We write P_+ for the set of dominant integral weights of $\mathfrak{g}$ and P_+^d for those of level d. For $\lambda \in P_+$, let $L(\lambda)$ denote the standard $\mathfrak{g}$-module of highest weight λ and write

$$L(\lambda) = \bigoplus_{\mu\in\mathfrak{h}^*} V_\mu$$

for its weight-space decomposition. The character of $L(\lambda)$ is then defined by

$$\operatorname{ch} L(\lambda) := \sum_{\mu\in\mathfrak{h}^*} \dim(V_\mu) e^\mu.$$

Moreover,

$$e^{-\lambda}\operatorname{ch} L(\lambda) \in \mathbb{Z}[[e^{-\alpha_0},\dots,e^{-\alpha_r}]],$$

since $\dim V_\mu = 0$ whenever $\lambda-\mu$ is not a nonnegative integral linear combination of simple roots.

Let W be the Weyl group of $\mathfrak{g}$, and let ρ be the Weyl vector, defined by $\langle \alpha_i^\vee, \rho\rangle = 1$ for $i = 0,1,\dots,r$. Then the Weyl–Kac character formula [20, 21] gives

$$\operatorname{ch} L(\lambda) = \frac{\sum_{w\in W} \operatorname{sgn}(w) e^{w(\lambda+\rho)-\rho}}{\prod_{\alpha>0}(1-e^{-\alpha})^{\operatorname{mult}(\alpha)}}.$$

The denominator product runs over the positive roots of $\mathfrak{g}$, with multiplicity $\operatorname{mult}(\alpha)$.

We call $e^{-\lambda}\operatorname{ch} L(\lambda)$ the modified character. Lepowsky and Wilson gave a Lie-theoretic interpretation and proof of the Rogers–Ramanujan identities [24–26, 28]. Lepowsky and Milne related the product side of the AGB identities to principal specializations of modified $A_1^{(1)}$ characters [27]. In the notation of (1.1), this relation is

$$e^{-\lambda}\operatorname{ch} L(\lambda)\big|_{(e^{-\alpha_0},e^{-\alpha_1})\mapsto(q,q)} = F\cdot\chi_{a,\sigma}^{k-1}(q), \tag{1.2}$$

where $F := (-q;q)_\infty = 1/(q;q^2)_\infty$ and $\lambda = (2k+\sigma-a)\Lambda_0 + a\Lambda_1 \in P_+^{2k+\sigma}$. Later, Meurman and Primc [32] gave a Lie–theoretic proof of all moduli of the Andrews–Gordon identities by studying higher-level standard modules of $A_1^{(1)}$.

Griffin, Ono and Warnaar [17] used a non-principal specialization of characters of $A_2^{(2)}$ standard modules. With the indexing of (1.1) and $\sigma = 1$, the rank-one specialization is

$$e^{-\lambda}\operatorname{ch} L(\lambda)\big|_{(e^{-\alpha_0},e^{-\alpha_1})\mapsto(-1,q)} = \chi_{a,1}^{k-1}(q), \qquad \lambda = 2(k-a)\Lambda_0 + a\Lambda_1 \in P_+^{2k}.$$

Here the fundamental weights have levels 1 and 2, respectively; compare [22, Section 1]. Their higher-rank identities are associated with $A_{2r}^{(2)}$ for $r>1$.

The Bailey lemma provides an analytic mechanism for producing multisums of AGB identities from an initial pair [7,39]. Andrews, Schilling and Warnaar introduced an A_2 Bailey lemma in [6], replacing the one-variable quadratic form s^2 by

$$r^2 - rs + s^2.$$

Warnaar's subsequent A_2 Bailey tree extends this idea by allowing several independent lattice directions, leading to all moduli of $A_2^{(1)}$ Rogers–Ramanujan-type identities [40]. Here and below, $\theta(z;q) = (z;q)_\infty (q/z;q)_\infty$, $\theta(z_1,\dots,z_m;q) = \prod_{i=1}^m \theta(z_i;q)$.

Theorem 1.1 (Warnaar). *Let $k \geq 1$ and $0 \leq a, b \leq k$ be integers, and let*

$$K_1 := 3k + \tau + 3, \qquad \tau \in \{-1, 0, 1\}.$$

Then

$$\sum_{\substack{\lambda_1 \geq \cdots \geq \lambda_k \geq 0 \\ \mu_1 \geq \cdots \geq \mu_k \geq 0}} \frac{1 - q^{\lambda_a + \mu_b + 1}}{1-q} \cdot \frac{q^{\sum_{i=1}^k (\lambda_i^2 - \lambda_i \mu_i + \mu_i^2) + \sum_{i=a+1}^k \lambda_i + \sum_{i=b+1}^k \mu_i}}{\prod_{i=1}^{k-1} (q;q)_{\lambda_i - \lambda_{i+1}} (q;q)_{\mu_i - \mu_{i+1}}}$$
$$\times \frac{q^{\tau(\tau-1)\lambda_k \mu_k}}{(q;q)_{\lambda_k + \mu_k} (q^2;q)_{\lambda_k + \mu_k}} \begin{bmatrix} \lambda_k + \mu_k \\ \lambda_k \end{bmatrix}_p = \frac{(q^{K_1};q^{K_1})_\infty^2}{(q;q)_\infty^3} \theta(q^{a+1}, q^{b+1}, q^{a+b+2}; q^{K_1}),$$

where $q^{\lambda_0} = q^{\mu_0} := 0$, $p = q$ if $\tau^2 = 1$, $p = q^3$ if $\tau = 0$.

This is Warnaar's proof of the Kanade–Russell conjecture [40, Theorem 1.2]. For $3 \nmid K_1$, multiplication of the common value by $q^{h_{a,b} - c/24}(q;q)_\infty$ gives a character $\chi_{a,b}^{K_1}(q)$ of the $W_3(3, K_1)$ vertex operator algebra, where $c = -2(K_1 - 4)(4K_1 - 9)/K_1$ [12, 40, 44]. Thus

$$\chi_{a,b}^{K_1}(q) = q^{h_{a,b} - c/24} \frac{(q^{K_1};q^{K_1})_\infty^2}{(q;q)_\infty^2} \theta(q^{a+1}, q^{b+1}, q^{a+b+2}; q^{K_1}),$$

where a, b, and K_1 are nonnegative integers such that $K_1 \geq 5$, $3 \nmid K_1$, and $a + b \leq K_1 - 3$, and where $h_{a,b} = (a^2 + ab + b^2 - (K_1 - 3)(a + b))/K_1$ is the conformal weight. Warnaar's A_2 Bailey tree also implies the A_2 Andrews–Gordon identities and their cylindric-partition interpretation [43].

On the sum side, the classical AGB identities involve a single family of variables λ, whereas the A_2 Bailey lemma involves two interacting families, λ and μ. On the product side, the corresponding theta products are naturally governed by A_1- and A_2-type Macdonald identities.

Another source of two-sequence multisums is the work of Capparelli, Meurman, Primc and Primc on coloured partition identities [9]. Russell obtained bivariate generating functions for their level-one cases and related them to identities of Jing, Misra, and Savage [38]. Kanade, Russell, Tsuchioka and Warnaar related these conjectures to non-standard specializations of standard modules for $A_{2n}^{(2)}$ and $D_{n+1}^{(2)}$, and formulated several Rogers–Ramanujan-type conjectures with two summation families. For $k \geq 1$, two of the conjectural sum–product identities discussed after [22, Conjecture 4.9] are

$$\sum_{\substack{r_1 \geq \cdots \geq r_k \geq 0 \\ s_k \geq \cdots \geq s_1 \geq 0}} \prod_{i=1}^k \frac{q^{(r_i+s_i)^2 + s_i^2 + s_i}}{(q;q)_{r_i - r_{i+1}} (q^2;q^2)_{s_i - s_{i-1}}} = \frac{(q, q^{k+1}, q^{k+2}; q^{k+2})_\infty}{(q;q^2)_\infty (q;q)_\infty}, \tag{1.3a}$$

$$\sum_{\substack{r_1 \geq \cdots \geq r_k \geq 0 \\ s_k \geq \cdots \geq s_1 \geq 0}} \prod_{i=1}^k \frac{q^{(r_i+s_i)^2 + s_i^2 + r_i + 2s_i}}{(q;q)_{r_i - r_{i+1}} (q^2;q^2)_{s_i - s_{i-1}}} = \frac{(q, q^2, q^{2k+2}, q^{2k+3}, q^{2k+4}, q^{2k+4}; q^{2k+4})_\infty}{(q^2;q^2)_\infty (q;q)_\infty}, \tag{1.3b}$$

where $r_{k+1} = s_0 = 0$. These identities are particularly close to the present work: their summands contain two interlaced variables r, s and $(q;q)$, $(q^2;q^2)$ factors respectively.

We use the following realisation of the B_2 root system [18]. Let

$$E = \mathbb{R}e_1 \oplus \mathbb{R}e_2$$

with standard inner product $(\cdot,\cdot) : E \times E \to \mathbb{R}$, and write the root system

$$\Phi(B_2) = \{\pm e_1, \pm e_2, \pm e_1 \pm e_2\}.$$

Then $(e_1, e_1) = (e_2, e_2) = 1$ and $(\pm e_1 \pm e_2, \pm e_1 \pm e_2) = 2$. Choose the short simple root $\alpha_1 = e_2$ and the long simple root $\alpha_2 = e_1 - e_2$. The corresponding Cartan matrix is

$$B = (a_{ij}), \quad a_{ij} = \frac{2(\alpha_i, \alpha_j)}{(\alpha_i, \alpha_i)}.$$

Then $B = \begin{pmatrix} 2 & -2 \\ -1 & 2 \end{pmatrix}$. To symmetrize the Cartan matrix, set $D := \mathrm{diag}(d_1, d_2)$, where $d_i = (\alpha_i, \alpha_i)$. Then

$$DB = \begin{pmatrix} 2 & -2 \\ -2 & 4 \end{pmatrix},$$

which is twice the Gram matrix of the simple roots The associated quadratic form is

$$x_1^2 - 2x_1x_2 + 2x_2^2 = \frac{1}{2}x^T DBx.$$

Equivalently, if

$$r\alpha_1 + \ell\alpha_2 = \ell e_1 + (r-\ell)e_2,$$

then

$$\|r\alpha_1 + \ell\alpha_2\|^2 = (r-\ell)^2 + \ell^2 = r^2 - 2r\ell + 2\ell^2,$$

which is the symmetrized B_2 quadratic form used below. In what follows, we shall refer to this symmetrized B_2 form simply as the B_2 form.

The purpose of this paper is to develop a Bailey-type mechanism governed by this symmetrized B_2 form. In contrast with the A_2 Bailey lemma, where the interaction is governed by $r^2 - rs + s^2$, the present construction distinguishes the two root lengths of B_2. This distinction is visible through the simultaneous appearance of $(q;q)$- and $(q^2;q^2)$-factors. The resulting chain and lattice may then be iterated from suitable Bailey pairs to produce Rogers–Ramanujan-type identities.

We write $\chi(A) = 1$ if the assertion A is true and $\chi(A) = 0$ otherwise. We can now state the first family of Rogers–Ramanujan-type identities associated with the symmetrized B_2 form.

Theorem 1.2. *Let $k \geq 1$ and $0 \leq s \leq k$ be integers, and let $z, w \in \mathbb{C}^\times$ satisfy $z^2 w = q^2$. Assume that $(zq, zwq; q)_\infty (wq^2; q^2)_\infty \neq 0$. Then*

$$\sum_{\substack{\lambda_1 \geq \cdots \geq \lambda_k \geq 0 \\ \mu_1 \geq \cdots \geq \mu_k \geq 0}} \frac{z^{\sum_{i=1}^k \lambda_i} w^{\sum_{i=1}^k \mu_i} q^{\sum_{i=1}^k (\lambda_i^2 - 2\lambda_i\mu_i + 2\mu_i^2)}}{\prod_{i=1}^{k-1} (q;q)_{\lambda_i - \lambda_{i+1}} (q^2;q^2)_{\mu_i - \mu_{i+1}}}$$
$$\times \frac{(zwq^{2\mu_k+1};q)_{\lambda_k} \left(q^{-S_s} - \chi(s \geq 1) z(1+w) q^{-S_{s-1}} + \chi(s \geq 2) z^2 w q^{-S_{s-2}}\right)}{(q, zq, zwq; q)_{\lambda_k} (q^2, wq^2; q^2)_{\mu_k}}$$
$$= \frac{(q^{2s+2}, q^{4k+4-2s}, q^{4k+6}; q^{4k+6})_\infty}{(zq, zwq; q)_\infty (wq^2; q^2)_\infty (q^2; q^2)_\infty}, \tag{1.4}$$

where $S_m = \lambda_1 + \cdots + \lambda_m$ for $m \geq 1$, and $S_0 = 0$. Terms with vanishing indicator are omitted, so that no negative-index partial sum is needed.

The right-hand side is a single A_1-type theta product. Let $B_2^{s,k}(z,q)$ denote the common value of the two sides of (1.4), with $w = q^2/z^2$. At $z = 1$, it gives a multisum expression for a principally specialized $A_1^{(1)}$-character.

Corollary 1.3. *Let $k \geq 1$ and $0 \leq a \leq k$ be integers, and set $\lambda = (2k+1-a)\Lambda_0 + a\Lambda_1 \in P_+^{2k+1}$. Then*

$$(q, q^3; q)_\infty (q^4; q^2)_\infty (-q^2; q^2)_\infty B_2^{a,k}(1, q) = e^{-\lambda} \operatorname{ch} L(\lambda)\big|_{(e^{-\alpha_0}, e^{-\alpha_1}) \mapsto (q^2, q^2)}.$$

For the second family, define, for nonnegative integers N, L,

$$\mathcal{R}_{N,L}(z,w) := \frac{(z^2 w q^{2N+1}; q^2)_L}{(q, zq; q)_N (q^2; q^2)_L (zwq; q)_{2L}}. \tag{1.5}$$

Actually, $\mathcal{R}_{N,L}(z,w)$ is the β-component of a B_2 Bailey pair $(\alpha_{u,v}, \beta_{u,v})$ constructed in Theorem 4.3. Iterating that pair gives the second family of identities.

Theorem 1.4. *Let $k \geq 1$ and $0 \leq s \leq k$ be integers, and let $z \in \mathbb{C}$ satisfy $z^2 = 1$. Then*

$$\begin{aligned} &\sum_{\substack{\lambda_1 \geq \cdots \geq \lambda_k \geq 0 \\ \mu_1 \geq \cdots \geq \mu_k \geq 0}} \frac{z^{\sum_{i=1}^k \lambda_i} q^{\sum_{i=1}^k (\lambda_i^2 - 2\lambda_i \mu_i + 2\mu_i^2) - \sum_{i=1}^s \lambda_i + 2\sum_{i=1}^k \mu_i}}{\prod_{i=1}^{k-1} (q;q)_{\lambda_i - \lambda_{i+1}} (q^2; q^2)_{\mu_i - \mu_{i+1}}} \mathcal{R}_{\lambda_k, \mu_k}(z, q^2) \\ &\qquad = \frac{(q^{4k+6}; q^{4k+6})_\infty^2}{(zq, zq^3; q)_\infty (q^2; q^2)_\infty^2} \sum_{r=0}^{s} z^r \theta(q^{2s-2r+2}, q^{2r+2}; q^{4k+6}). \end{aligned} \tag{1.6}$$

The finite convolution of two A_1-theta products on the right-hand side of (1.6) comes from the character expansion proved in Proposition 4.4. The corresponding finite-dimensional branching rule is

$$V_{B_2}(s\omega_{\mathrm{sp}}) \downarrow_{D_2 = A_1 \oplus A_1} \cong \bigoplus_{r=0}^{s} U_{s-r} \boxtimes U_r,$$

where $U_m \cong \mathrm{Sym}^m(\mathbb{C}^2)$ is the irreducible A_1-module of highest weight m. Section 4.3 explains how its normalized character enters the two theta evaluations.

The following special cases are obtained by setting $s = 0$ in (1.4), and by setting $z = 1$ and $s = 0, 1$ in (1.6).

$$\begin{aligned} &\sum_{\substack{\lambda_1 \geq \cdots \geq \lambda_k \geq 0 \\ \mu_1 \geq \cdots \geq \mu_k \geq 0}} \frac{z^{\sum_{i=1}^k \lambda_i} w^{\sum_{i=1}^k \mu_i} q^{\sum_{i=1}^k (\lambda_i^2 - 2\lambda_i \mu_i + 2\mu_i^2)}}{\prod_{i=1}^{k-1} (q;q)_{\lambda_i - \lambda_{i+1}} (q^2; q^2)_{\mu_i - \mu_{i+1}}} \frac{(zwq^{2\mu_k+1}; q)_{\lambda_k}}{(q, zq, zwq; q)_{\lambda_k} (q^2, wq^2; q^2)_{\mu_k}} \\ &= \frac{(q^2, q^{4k+4}, q^{4k+6}; q^{4k+6})_\infty}{(zq, zwq; q)_\infty (wq^2; q^2)_\infty (q^2; q^2)_\infty}, \qquad \text{for } z^2 w = q^2, \end{aligned} \tag{1.7a}$$

$$\begin{aligned} &\sum_{\substack{\lambda_1 \geq \cdots \geq \lambda_k \geq 0 \\ \mu_1 \geq \cdots \geq \mu_k \geq 0}} \frac{q^{\sum_{i=1}^k (\lambda_i^2 - 2\lambda_i \mu_i + 2\mu_i^2) + 2\sum_{i=1}^k \mu_i - \sum_{i=1}^\sigma \lambda_i}}{\prod_{i=1}^{k-1} (q;q)_{\lambda_i - \lambda_{i+1}} (q^2; q^2)_{\mu_i - \mu_{i+1}}} \frac{(q^{2\mu_k+3}; q)_{\lambda_k} (q^{2\lambda_k+3}; q^2)_{\mu_k}}{(q, q, q^3; q)_{\lambda_k} (q^2; q^2)_{\mu_k} (q^{\lambda_k+3}; q)_{2\mu_k}} \\ &= \frac{(1+\sigma)(q^2, q^{4k+4}, q^{4k+6}, q^{2\sigma+2}, q^{4k+4-2\sigma}, q^{4k+6}; q^{4k+6})_\infty}{(q, q^3; q)_\infty (q^2; q^2)_\infty^2}, \end{aligned} \tag{1.7b}$$

where $\sigma \in \{0, 1\}$. To some extent, (1.3a), (1.3b), (1.7a), and (1.7b) may be viewed as identities of the same general type. The right-hand sides of (1.3a) and (1.7a) contain a single theta function. In contrast, (1.3b) and (1.7b) are naturally related to the $D_2^{(1)}$ Macdonald identity. Appendix A gives a related projection of Warnaar's A_2 Bailey tree.

The paper is organized as follows. Section 2 introduces the notation and summation formulas. Section 3 establishes the B_2 Bailey lemma and the chain and lattice transformations. Section 4 constructs two initial Bailey pairs and identifies the polynomial needed for their limits with a normalized spinor character. Section 5 proves the two main families of Rogers–Ramanujan-type identities. Section 6 derives the character, CMPP partition, and Hall–Littlewood applications. Section 7 discusses further directions, and Appendix A proves the projected A_2 identity.

2 Preliminaries

We use the standard q-series notation of [14] and assume throughout that $0 < |q| < 1$. The q-shifted factorial is defined by

$$(a)_n = (a;q)_n = \begin{cases} 1, & \text{if } n = 0, \\ (1-a)(1-aq)\cdots(1-aq^{n-1}), & \text{if } n \geq 1, \end{cases}$$

We adopt the convention $1/(q;q)_n = 0$ for $n < 0$, and the analogous convention with base q^2. We also write

$$(a)_\infty = (a;q)_\infty = \prod_{n=0}^{\infty}(1-aq^n).$$

For products of q-shifted factorials, we use

$$(a_1, a_2, \ldots, a_m; q)_n = (a_1;q)_n (a_2;q)_n \cdots (a_m;q)_n,$$

for $n \in \mathbb{Z}_{\geq 0} \cup \{\infty\}$. For negative indices, we define

$$(a;q)_{-n} := \frac{1}{(aq^{-n};q)_n} = \frac{(-q/a)^n q^{\binom{n}{2}}}{(q/a;q)_n}, \quad n \in \mathbb{Z}_{>0}.$$

The Gaussian polynomials, or q-binomial coefficients, are defined for integers $n \geq 0$ and k by

$$\begin{bmatrix} n \\ k \end{bmatrix} = \begin{cases} \dfrac{(q;q)_n}{(q;q)_k (q;q)_{n-k}}, & \text{if } 0 \leq k \leq n, \\ 0, & \text{otherwise,} \end{cases}$$

We write $\begin{bmatrix} n \\ k \end{bmatrix}_{q^t}$ for the same expression with q replaced by q^t.

Recall that the basic hypergeometric series ${}_r\phi_s$ is defined by [14]

$${}_r\phi_s\left(\begin{matrix} a_1, a_2, \ldots, a_r \\ b_1, b_2, \ldots, b_s \end{matrix}; q, x\right) = \sum_{n=0}^{\infty} \frac{(a_1, a_2, \ldots, a_r; q)_n}{(q, b_1, \ldots, b_s; q)_n} \left[(-1)^n q^{\binom{n}{2}}\right]^{1+s-r} x^n.$$

The bilateral basic hypergeometric series is defined by

$${}_r\psi_s\left(\begin{matrix} a_1, a_2, \ldots, a_r \\ b_1, b_2, \ldots, b_s \end{matrix}; q, x\right) = \sum_{n=-\infty}^{\infty} \frac{(a_1, a_2, \ldots, a_r; q)_n}{(b_1, \ldots, b_s; q)_n} \left[(-1)^n q^{\binom{n}{2}}\right]^{s-r} x^n.$$

We use the following formulas from [14]. For $|de/(abc)| < 1$ and $|e/a| < 1$, a transformation for ${}_3\phi_2$ is

$${}_3\phi_2\left[\begin{matrix} a, b, c \\ d, e \end{matrix}; q, \frac{de}{abc}\right] = \frac{(e/a, de/bc; q)_\infty}{(e, de/abc; q)_\infty} {}_3\phi_2\left[\begin{matrix} a, d/b, d/c \\ d, de/bc \end{matrix}; q, \frac{e}{a}\right]. \tag{2.1}$$

The terminating very-well-poised ${}_6\phi_5$ summation, for integers $k \geq 0$ is

$$ {}_6\phi_5\left[\begin{matrix} a,\ q\sqrt{a},\ -q\sqrt{a},\ b,\ c,\ q^{-k} \\ \sqrt{a},\ -\sqrt{a},\ aq/b,\ aq/c,\ aq^{k+1}\end{matrix}; q, \frac{aq^{k+1}}{bc}\right] = \frac{(aq, aq/bc; q)_k}{(aq/b, aq/c; q)_k}. \tag{2.2} $$

Bailey's very-well-poised ${}_6\psi_6$ summation for $|qa^2/(bcde)| < 1$ is

$$ \begin{aligned} &{}_6\psi_6\left[\begin{matrix} qa^{\frac{1}{2}}, & -qa^{\frac{1}{2}}, & b, & c, & d, & e, \\ a^{\frac{1}{2}}, & -a^{\frac{1}{2}}, & aq/b, & aq/c, & aq/d, & aq/e\end{matrix}; q, \frac{qa^2}{bcde}\right] \\ &= \frac{(aq, aq/bc, aq/bd, aq/be, aq/cd, aq/ce, aq/de, q, q/a; q)_\infty}{(aq/b, aq/c, aq/d, aq/e, q/b, q/c, q/d, q/e, qa^2/bcde; q)_\infty}. \end{aligned} \tag{2.3} $$

We also use the standard abbreviation

$$ {}_6W_5(a; b, c, d; q, z) = {}_6\phi_5\left[\begin{matrix} a, q\sqrt{a}, -q\sqrt{a}, b, c, d \\ \sqrt{a}, -\sqrt{a}, aq/b, aq/c, aq/d\end{matrix}; q, z\right]. $$

3 A symmetrized B_2 Bailey lemma

3.1 The B_2 Bailey chain

We begin with the finite identity underlying the Bailey transformation. For integers $0 \leq r \leq N$ and $0 \leq \ell \leq L$, define the kernel

$$ \mathcal{K}_{N,L;r,\ell}(z, w; q) := \frac{z^r w^\ell q^{r^2-2r\ell+2\ell^2}}{(q; q)_{N-r}(q^2; q^2)_{L-\ell}}. \tag{3.1} $$

The exponent is the symmetrized B_2 quadratic form, while the two factorial bases are compatible with the two root lengths: the short-root direction is measured in steps of q, and the long-root direction in steps of q^2. For a function F of two nonnegative integer variables, define the operator

$$ \mathscr{K}_{N,L}(z, w) \cdot F_{r,\ell} := \sum_{r=0}^{N}\sum_{\ell=0}^{L} \mathcal{K}_{N,L;r,\ell}(z, w; q) F_{r,\ell}. \tag{3.2} $$

The construction is based on the following finite invariant of the kernel operator:

$$ \Psi_{N,L}(z, w; q) = \frac{1}{(zq; q)_N (zwq; q)_{2L}} \sum_{j=0}^{\min(\lfloor N/2\rfloor, L)} \frac{(z^2w)^j q^{2j^2}(zwq^{N+1}, zwq^{N+2}; q^2)_{L-j}}{(q; q)_{N-2j}(q^2, z^2wq^2; q^2)_j (q^2, wq^2; q^2)_{L-j}}. \tag{3.3} $$

We set $\Psi_{N,L}(z, w; q) = 0$ if $N < 0$ or $L < 0$. For comparison, Warnaar's A_2 construction uses the rational function [40, 41]

$$ \Phi_{n,m}(z, w; q) := \frac{(zwq; q)_{n+m}}{(q, zq, zwq; q)_n (q, wq, zwq; q)_m}, $$

and the associated kernel

$$ \frac{z^r w^s q^{r^2-rs+s^2}}{(q; q)_{n-r}(q; q)_{m-s}}. $$

We use the following decomposition of $\Psi_{N,L}(z, w; q)$:

$$ \Psi_{N,L}(z, w; q) = \sum_{j\geq 0} c_j(z, w; q) \mathcal{A}_{N-2j,L-j}(zq^{2j}, w; q). \tag{3.4} $$

Here the coefficients $c_j(z,w;q)$ and the product block $\mathcal{A}_{N,L}(z,w;q)$ are defined by

$$c_j(z,w;q) := \frac{(z^2w)^j q^{2j^2}}{(q^2;q^2)_j (z^2wq^2;q^2)_j (zq,zwq;q)_{2j}}, \tag{3.5}$$

$$\mathcal{A}_{N,L}(z,w;q) := \frac{(zwq^{2L+1};q)_N}{(q,zq,zwq;q)_N (q^2,wq^2;q^2)_L}. \tag{3.6}$$

We set $\mathcal{A}_{N,L}(z,w;q)=0$ if $N<0$ or $L<0$.

The basic invariance property is the following finite identity. It is the B_2 analogue of the elementary Bailey chain identity.

Theorem 3.1. *For all nonnegative integers N, L,*

$$\sum_{r=0}^{N}\sum_{\ell=0}^{L} \frac{z^r w^\ell q^{r^2-2r\ell+2\ell^2}}{(q;q)_{N-r}(q^2;q^2)_{L-\ell}} \Psi_{r,\ell}(z,w;q) = \Psi_{N,L}(z,w;q). \tag{3.7}$$

The proof has two steps, recorded in the next two lemmas. We first evaluate the action of $\mathscr{K}_{N,L}(z,w)$ on $\mathcal{A}_{N,L}$. Then we show that the coefficients c_j satisfy the convolution identity needed to recover $\Psi_{N,L}$.

Lemma 3.2. *Let*

$$d_h(z,w;q) := \frac{(z^2w)^h q^{2h^2}}{(q^2;q^2)_h (zq,zwq;q)_{2h}}. \tag{3.8}$$

Then, for nonnegative integers N, L,

$$\mathscr{K}_{N,L}(z,w)\cdot \mathcal{A}_{r,\ell} = \sum_{h\ge 0} d_h(z,w;q)\, \mathcal{A}_{N-2h,L-h}(zq^{2h},w;q). \tag{3.9}$$

Proof. By definition,

$$\mathscr{K}_{N,L}(z,w)\cdot \mathcal{A}_{r,\ell} = \sum_{r=0}^{N}\sum_{\ell=0}^{L} \frac{z^r w^\ell q^{r^2-2r\ell+2\ell^2}}{(q;q)_{N-r}(q^2;q^2)_{L-\ell}} \frac{(zwq^{2\ell+1};q)_r}{(q,zq,zwq;q)_r (q^2,wq^2;q^2)_\ell}.$$

Write the inner r-sum as

$$\frac{1}{(q;q)_N} {}_2\phi_2\left[\begin{matrix} zwq^{2\ell+1},\ q^{-N} \\ zwq,\ zq \end{matrix}; q,\ zq^{N-2\ell+1}\right].$$

We use the following limiting case of the terminating transformation [14, III.9]

$${}_2\phi_2\left[\begin{matrix} a,\ q^{-n} \\ b,\ c \end{matrix}; q, \frac{bcq^n}{a}\right] = \frac{1}{(c;q)_n} {}_2\phi_1\left[\begin{matrix} b/a,\ q^{-n} \\ b \end{matrix}; q, cq^n\right],$$

with

$$a = zwq^{2\ell+1}, \qquad b = zwq, \qquad c = zq, \qquad n = N.$$

This gives

$$\mathscr{K}_{N,L}(z,w)\cdot \mathcal{A}_{r,\ell} = \frac{1}{(q;q)_N (zq;q)_N} S_{N,L}, \tag{3.10}$$

where

$$S_{N,L} := \sum_{\ell=0}^{L} \frac{w^\ell q^{2\ell^2}}{(q^2;q^2)_{L-\ell}(q^2,wq^2;q^2)_\ell} {}_2\phi_1\left[\begin{matrix} q^{-2\ell},\ q^{-N} \\ zwq \end{matrix}; q, zq^{N+1}\right]. \tag{3.11}$$

It remains to evaluate $S_{N,L}$ and we claim that

$$S_{N,L} = \frac{(q;q)_N}{(zwq;q)_N} \sum_{h=0}^{\min(\lfloor N/2 \rfloor, L)} \frac{(z^2w)^h q^{2h^2} (zwq^{2L+1};q)_{N-2h}}{(q^2;q^2)_h (q;q)_{N-2h} (q^2, wq^2;q^2)_{L-h}}. \tag{3.12}$$

To prove this claim, multiply both sides by $(zwq;q)_N/(q;q)_N$. Both resulting expressions are polynomials in z of degree at most N, so it suffices to compare their coefficients of z^n for $0 \le n \le N$. Expand the ${}_2\phi_1$ in (3.11) and use

$$(zwq^{p+1};q)_{N-p} = \sum_{r=0}^{N-p} (-1)^r q^{\binom{r}{2}} \begin{bmatrix} N-p \\ r \end{bmatrix}_q (zwq^{p+1})^r,$$

to obtain

$$\begin{aligned}
&[z^n] \frac{(zwq;q)_N}{(q;q)_N} S_{N,L} \\
&= \sum_{\ell=0}^{L} \frac{w^\ell q^{2\ell^2}}{(q^2;q^2)_{L-\ell} (q^2, wq^2;q^2)_\ell} \sum_{p=0}^{n} \frac{(q^{-2\ell};q)_p}{(q;q)_p} \frac{(-1)^p q^{\binom{p+1}{2}}}{(q;q)_{N-p}} \\
&\quad \times (-1)^{n-p} q^{\binom{n-p}{2}} \begin{bmatrix} N-p \\ n-p \end{bmatrix}_q w^{n-p} q^{(p+1)(n-p)} \\
&= \frac{(-1)^n q^{\binom{n+1}{2}}}{(q;q)_{N-n}} C_{n,L},
\end{aligned} \tag{3.13}$$

where

$$A_\ell := \frac{w^\ell q^{2\ell^2}}{(q^2;q^2)_{L-\ell} (q^2, wq^2;q^2)_\ell}, \qquad C_{n,L} := \sum_{\ell=0}^{L} A_\ell \sum_{p=0}^{n} \frac{w^{n-p} (q^{-2\ell};q)_p}{(q;q)_p (q;q)_{n-p}}.$$

Define $C_{n,L}$ by the displayed sum for every $n \ge 0$. Taking $n = p + r$, its generating function is

$$\begin{aligned}
\sum_{n \ge 0} C_{n,L} x^n &= \sum_{\ell=0}^{L} A_\ell \sum_{p,r \ge 0} \frac{(q^{-2\ell};q)_p}{(q;q)_p} \frac{w^r}{(q;q)_r} x^{p+r} \\
&= \sum_{\ell=0}^{L} A_\ell \Big(\sum_{p \ge 0} \frac{(q^{-2\ell};q)_p}{(q;q)_p} x^p \Big) \Big(\sum_{r \ge 0} \frac{(wx)^r}{(q;q)_r} \Big) \\
&= \frac{1}{(wx;q)_\infty} \sum_{\ell=0}^{L} \frac{w^\ell q^{2\ell^2} (q^{-2\ell}x;q)_{2\ell}}{(q^2;q^2)_{L-\ell} (q^2, wq^2;q^2)_\ell}.
\end{aligned} \tag{3.14}$$

Here we have used the q-binomial theorem [14, II.1]. These generating-function calculations are valid as formal power series in x; alternatively, one may first take x sufficiently small. The sum

$$\sum_{\ell=0}^{L} \frac{w^\ell q^{2\ell^2} (q^{-2\ell}x;q)_{2\ell}}{(q^2;q^2)_{L-\ell} (q^2, wq^2;q^2)_\ell} = \sum_{h=0}^{L} \frac{w^h q^{-h} x^{2h} (wx;q)_{2L-2h}}{(q^2;q^2)_h (q^2, wq^2;q^2)_{L-h}} \tag{3.15}$$

follows from a terminating ${}_3\phi_1$ transformation. To make the specialization explicit, put $Q = q^2$. Specializing (2.1) to $a = Q^{-L}$ first gives the finite prefactor

$$\frac{(eQ^L, de/(bc);Q)_\infty}{(e, deQ^L/(bc);Q)_\infty} = \frac{(de/(bc);Q)_L}{(e;Q)_L}.$$

After this cancellation, both series terminate. Taking $e \to \infty$ therefore yields

$$ {}_3\phi_1\left[\begin{matrix} Q^{-L}, b, c \\ d \end{matrix}; Q, \frac{dQ^L}{bc}\right] = \left(\frac{d}{bc}\right)^L {}_3\phi_1\left[\begin{matrix} Q^{-L}, d/b, d/c \\ d \end{matrix}; Q, \frac{bcQ^L}{d}\right]. $$

Now take $b = q/x$, $c = q^2/x$, and $d = wq^2$. The left-hand side of (3.15) is

$$ \frac{1}{(Q;Q)_L} {}_3\phi_1\left[\begin{matrix} Q^{-L}, q/x, q^2/x \\ wq^2 \end{matrix}; Q, wx^2q^{2L-1}\right]. $$

Applying the transformation and replacing its summation index by $L-h$ gives the right-hand side of (3.15). Since both sides are polynomials in x, the identity also holds at $x = 0$. Substituting (3.15) into (3.14) and expanding $1/(wxq^{2L-2h};q)_\infty$ by the q-binomial theorem [14, II.1] yields

$$ C_{n,L} = \sum_{h=0}^{\min(\lfloor n/2 \rfloor, L)} \frac{w^{n-h} q^{(2L-2h)(n-2h)-h}}{(q^2;q^2)_h (q;q)_{n-2h} (q^2, wq^2; q^2)_{L-h}}. \tag{3.16} $$

On the other hand, the coefficient of z^n on the right-hand side of (3.12), after multiplication by $(zwq;q)_N/(q;q)_N$, is exactly

$$ \frac{(-1)^n q^{\binom{n+1}{2}}}{(q;q)_{N-n}} \sum_{h=0}^{\min(\lfloor n/2 \rfloor, L)} \frac{w^{n-h} q^{(2L-2h)(n-2h)-h}}{(q^2;q^2)_h (q;q)_{n-2h} (q^2, wq^2; q^2)_{L-h}}, $$

since the normalized right-hand side expands as

$$ \sum_{h=0}^{\min(\lfloor N/2 \rfloor, L)} \sum_{r=0}^{N-2h} \frac{(-1)^r z^{2h+r} w^{h+r} q^{2h^2 + \binom{r}{2} + (2L+1)r}}{(q^2;q^2)_h (q;q)_r (q;q)_{N-2h-r} (q^2, wq^2; q^2)_{L-h}} $$

after expanding $(zwq^{2L+1};q)_{N-2h}/(q;q)_{N-2h}$ and setting $r = n - 2h$. In simplifying the exponent, we use

$$ 2h^2 + \binom{n-2h}{2} + (2L+1)(n-2h) = \binom{n+1}{2} + (2L-2h)(n-2h) - h. $$

This agrees with (3.13) by (3.16). Hence our claim (3.12) follows.

Combining (3.10) and (3.12), we obtain

$$ \mathscr{K}_{N,L}(z,w) \cdot \mathcal{A}_{r,\ell} = \frac{1}{(zq, zwq; q)_N} \sum_{h=0}^{\min(\lfloor N/2 \rfloor, L)} \frac{(z^2w)^h q^{2h^2} (zwq^{2L+1};q)_{N-2h}}{(q^2;q^2)_h (q;q)_{N-2h} (q^2, wq^2; q^2)_{L-h}}. \tag{3.17} $$

Using $(zq, zwq; q)_{2h} (zq^{2h+1}, zwq^{2h+1}; q)_{N-2h} = (zq, zwq; q)_N$, we identify the h-th summand of (3.17), including its prefactor, as

$$ d_h(z,w;q) \mathcal{A}_{N-2h, L-h}(zq^{2h}, w; q). $$

This proves (3.9). □

Lemma 3.3. *For every nonnegative integer k,*

$$ \sum_{j=0}^{k} c_j(z,w;q) (z^2w)^j q^{2j^2} d_{k-j}(zq^{2j}, w; q) = c_k(z,w;q). \tag{3.18} $$

Proof. Let $a = z^2 w$. Using definitions (3.5) and (3.8), together with

$$(zq, zwq; q)_{2j}(zq^{2j+1}, zwq^{2j+1}; q)_{2k-2j} = (zq, zwq; q)_{2k},$$

the j-th summand on the left of (3.18) becomes

$$\frac{a^k q^{2k^2}}{(zq, zwq; q)_{2k}} \frac{a^j q^{2j^2}}{(q^2; q^2)_j (aq^2; q^2)_j (q^2; q^2)_{k-j}}.$$

Thus (3.18) is equivalent to

$$\sum_{j=0}^{k} \frac{a^j q^{2j^2}}{(q^2; q^2)_j (aq^2; q^2)_j (q^2; q^2)_{k-j}} = \frac{1}{(q^2; q^2)_k (aq^2; q^2)_k}.$$

For completeness, put $Q = q^2$ and apply the q-Chu–Vandermonde summation [14, II.7] in the form

$${}_2\phi_1 \left[\begin{matrix} Q^{-k}, b \\ aQ \end{matrix}; Q, \frac{aQ^{k+1}}{b} \right] = \frac{(aQ/b; Q)_k}{(aQ; Q)_k}.$$

Taking $b \to \infty$ in this finite sum proves (3.18). □

Proof of Theorem 3.1. For $0 \le j \le \min(\lfloor N/2 \rfloor, L)$, put $F_{r,\ell}^{(j)} := \mathcal{A}_{r-2j,\ell-j}(zq^{2j}, w; q)$. Substitution of (3.4) gives

$$\mathscr{K}_{N,L}(z, w) \cdot \Psi_{r,\ell} = \sum_{j \ge 0} c_j(z, w; q) \mathscr{K}_{N,L}(z, w) \cdot F_{r,\ell}^{(j)}.$$

For fixed j, only terms with $r \ge 2j$ and $\ell \ge j$ contribute. Put $r = R + 2j$ and $\ell = S + j$ in $\mathcal{K}$. Then

$$r^2 - 2r\ell + 2\ell^2 = R^2 - 2RS + 2S^2 + 2jR + 2j^2,$$

and

$$z^r w^\ell q^{r^2 - 2r\ell + 2\ell^2} = (z^2 w)^j q^{2j^2} (zq^{2j})^R w^S q^{R^2 - 2RS + 2S^2}.$$

Consequently, for each fixed j,

$$\mathscr{K}_{N,L}(z, w) \cdot F_{r,\ell}^{(j)} = (z^2 w)^j q^{2j^2} \mathscr{K}_{N-2j,L-j}(zq^{2j}, w) \cdot \mathcal{A}_{R,S}(zq^{2j}, w; q), \tag{3.19}$$

where $0 \le R \le N - 2j$ and $0 \le S \le L - j$ are the variables of the sum on the right. Applying Lemma 3.2 with (N, L, z, w) replaced by $(N - 2j, L - j, zq^{2j}, w)$, and then summing over j, gives

$$\sum_{j,h \ge 0} c_j(z, w; q)(z^2 w)^j q^{2j^2} d_h(zq^{2j}, w; q) \mathcal{A}_{N-2j-2h,L-j-h}(zq^{2j+2h}, w; q).$$

Put $k = j + h$. The coefficient of $\mathcal{A}_{N-2k,L-k}(zq^{2k}, w; q)$ is the left side of (3.18), hence equals $c_k(z, w; q)$. Therefore

$$\mathscr{K}_{N,L}(z, w) \cdot \Psi_{r,\ell} = \sum_{k \ge 0} c_k(z, w; q) \mathcal{A}_{N-2k,L-k}(zq^{2k}, w; q) = \Psi_{N,L}(z, w; q).$$

□

3.2 The B_2 Bailey lattice

Theorem 3.1 gives a chain of iterations with fixed parameters. A parameter shift provides a second direction of iteration and leads to a lattice. We first establish a four-term relation for $\Psi_{N,L}(z,w)$, then use it to define a deformation $\Psi_{N,L}(U,V;z,w)$. For brevity, write $\Psi_{N,L}(z,w)=\Psi_{N,L}(z,w;q)$.

Lemma 3.4. *Let N,L be nonnegative integers. Then*

$$\Psi_{N,L}(z,w)=\Psi_{N,L}(z/q,w)-\frac{z}{(z;q)_2}\Psi_{N-1,L}(zq,wq^{-2}) \tag{3.20}$$
$$+\frac{zw^2}{(w;q^2)_2(zw;q)_2}\Psi_{N-1,L-1}(z/q,wq^2)-\frac{z^4w^2}{(z;q)_2(zw;q)_2(z^2w;q^2)_2}\Psi_{N-2,L-1}(zq,w).$$

Proof. Write

$$\Psi_{N,L}(z,w;q)=\sum_{j\geq 0}\psi_j(N,L;z,w;q),$$

where ψ_j is the j-th summand in (3.3), with $\psi_j(N,L;z,w;q)=0$ unless $N,L\geq 0$ and $0\leq j\leq \min(\lfloor N/2\rfloor,L)$. For brevity, set $\psi_j=\psi_j(N,L;z,w;q)$. After shifting the index in the last sum of (3.20), define

$$\Delta_j=-\psi_j(N,L;z,w)+\psi_j(N,L;z/q,w)+\frac{zw^2}{(w;q^2)_2(zw;q)_2}\psi_j(N-1,L-1;z/q,wq^2)$$
$$-\frac{z}{(z;q)_2}\psi_j(N-1,L;zq,wq^{-2})-\frac{z^4w^2}{(z;q)_2(zw;q)_2(z^2w;q^2)_2}\psi_{j-1}(N-2,L-1;zq,w).$$

It remains to show that $\sum_j\Delta_j=0$. Note that

$$\Delta_j=Y_j\psi_j(N,L;z,w)-Y_{j+1}\psi_{j+1}(N,L;z,w),$$

where

$$Y_j=\frac{q^{-2j}(1-q^{2j})(1-z^2wq^{2j})}{(1-z)(1-zw)}.$$

Write $m=N-2j$ and $r=L-j$. Direct calculations give:

$$\frac{\psi_j(N,L;z/q,w)}{\psi_j}=q^{-2j}\frac{(1-z^2wq^{2j})(1-zq^N)(1-zwq^N)(1-zwq^{2L})}{(1-z^2w)(1-z)(1-zw)(1-zwq^{2L+m})},$$
$$\frac{z}{(z;q)_2}\frac{\psi_j(N-1,L;zq,wq^{-2})}{\psi_j}=\frac{z(1-q^m)(1-wq^{2r})(1-zwq^{2L})(1-zwq^{N-1})(1-zwq^N)}{(1-w)(1-z)(1-zw)(1-zwq^{2L+m-1})(1-zwq^{2L+m})},$$
$$\frac{zw^2}{(w;q^2)_2(zw;q)_2}\frac{\psi_j(N-1,L-1;z/q,wq^2)}{\psi_j}$$
$$=\frac{zw^2(1-q^m)(1-q^{2r})(1-zwq^{2L})(1-zq^{N-1})(1-zq^N)}{(1-w)(1-z)(1-zw)(1-zwq^{2L+m-1})(1-zwq^{2L+m})},$$
$$\frac{z^4w^2}{(z;q)_2(zw;q)_2(z^2w;q^2)_2}\frac{\psi_{j-1}(N-2,L-1;zq,w)}{\psi_j}$$
$$=\frac{z^2wq^{-2j}(1-q^{2j})(1-zwq^{2L})(1-zq^N)(1-zwq^N)}{(1-z)(1-zw)(1-z^2w)(1-zwq^{2L+m})}.$$

On the other hand,

$$\frac{\psi_{j+1}}{\psi_j}=\frac{z^2wq^{4j+2}(1-q^{m-1})(1-q^m)(1-q^{2r})(1-wq^{2r})}{(1-q^{2j+2})(1-z^2wq^{2j+2})(1-zwq^{2L+m-1})(1-zwq^{2L+m})}.$$

Thus

$$Y_{j+1}\frac{\psi_{j+1}}{\psi_j}=\frac{z^2wq^{2j}(1-q^{m-1})(1-q^m)(1-q^{2r})(1-wq^{2r})}{(1-z)(1-zw)(1-zwq^{2L+m-1})(1-zwq^{2L+m})}.$$

Substituting these ratios into Δ_j/ψ_j and combining over a common denominator exactly yields

$$\frac{\Delta_j}{\psi_j}=Y_j-Y_{j+1}\frac{\psi_{j+1}}{\psi_j}.$$

Now put $M=\min(\lfloor N/2\rfloor,L)$. The telescoping sum is

$$\sum_{j=0}^{M}\Delta_j=\sum_{j=0}^{M}(Y_j\psi_j-Y_{j+1}\psi_{j+1})=Y_0\psi_0-Y_{M+1}\psi_{M+1}.$$

Since $Y_0=0$ and $\psi_{M+1}(N,L;z,w)=0$, the telescoping sum vanishes. This proves the lemma. □

The lemma motivates the following two-parameter deformation:

$$\begin{aligned}\Psi_{N,L}(U,V;z,w;q):=\Psi_{N,L}(z/q,w)-\frac{Uz}{(z;q)_2}\Psi_{N-1,L}(zq,wq^{-2})\\+\frac{UVzw^2}{(w;q^2)_2(zw;q)_2}\Psi_{N-1,L-1}(z/q,wq^2)-\frac{U^2Vz^4w^2}{(z;q)_2(zw;q)_2(z^2w;q^2)_2}\Psi_{N-2,L-1}(zq,w).\end{aligned}\tag{3.21}$$

It satisfies

$$\Psi_{N,L}(1,1;z,w)=\Psi_{N,L}(z,w).$$

We also set $\Psi_{N,L}(U,V;z,w;q)=0$ if $N<0$ or $L<0$.

The deformed function satisfies the following chain and lattice identities.

Theorem 3.5. *For nonnegative integers N,L and arbitrary parameters U,V,*

$$\sum_{r=0}^{N}\sum_{\ell=0}^{L}\mathcal{K}_{N,L;r,\ell}(z,w;q)\Psi_{r,\ell}(1,1;z,w;q)=\Psi_{N,L}(1,1;z,w;q)\tag{3.22a}$$

and

$$\sum_{r=0}^{N}\sum_{\ell=0}^{L}\mathcal{K}_{N,L;r,\ell}(z/q,w;q)\Psi_{r,\ell}(U,V;z,w;q)=\Psi_{N,L}\big(Uz,Vw;z,w;q\big).\tag{3.22b}$$

Proof. Identity (3.22a) is Theorem 3.1, since $\Psi_{N,L}(1,1;z,w;q)=\Psi_{N,L}(z,w;q)$. To prove (3.22b), it suffices to compare the coefficients of $1,U,UV,U^2V$, the only monomials present on either side. By the definition of $\mathcal{K}_{N,L;r,\ell}$, we have the following kernel-shift identities:

$$\begin{aligned}\mathcal{K}_{N,L;r+1,\ell}(z/q,w;q)&=z\mathcal{K}_{N-1,L;r,\ell}(zq,wq^{-2};q),\\\mathcal{K}_{N,L;r+1,\ell+1}(z/q,w;q)&=zw\mathcal{K}_{N-1,L-1;r,\ell}(z/q,wq^2;q),\\\mathcal{K}_{N,L;r+2,\ell+1}(z/q,w;q)&=z^2w\mathcal{K}_{N-2,L-1;r,\ell}(zq,w;q).\end{aligned}\tag{3.23}$$

We begin with the constant coefficient of the left-hand side of (3.22b):

$$\sum_{r=0}^{N}\sum_{\ell=0}^{L}\mathcal{K}_{N,L;r,\ell}(z/q,w;q)\Psi_{r,\ell}(z/q,w;q).$$

Apply Theorem 3.1 with $(z,w)\mapsto(z/q,w)$. Then

$$\sum_{r=0}^{N}\sum_{\ell=0}^{L}\mathcal{K}_{N,L;r,\ell}(z/q,w;q)\Psi_{r,\ell}(z/q,w;q)=\Psi_{N,L}(z/q,w;q),$$

which is exactly the constant term of the right-hand side:

$$\Psi_{N,L}(Uz,Vw;z,w;q).$$

The coefficient of U on the left-hand side of (3.22b) is

$$-\frac{z}{(z;q)_2}\sum_{r=0}^{N}\sum_{\ell=0}^{L}\mathcal{K}_{N,L;r,\ell}(z/q,w;q)\Psi_{r-1,\ell}(zq,wq^{-2};q).$$

Since $\Psi_{r-1,\ell}=0$ if $r=0$, we may replace the old summation index by $r+1$. Thus this coefficient equals

$$-\frac{z}{(z;q)_2}\sum_{r=0}^{N-1}\sum_{\ell=0}^{L}\mathcal{K}_{N,L;r+1,\ell}(z/q,w;q)\Psi_{r,\ell}(zq,wq^{-2};q).$$

Using the kernel ratio $\mathcal{K}_{N,L;r+1,\ell}(z/q,w;q)=z\mathcal{K}_{N-1,L;r,\ell}(zq,wq^{-2};q)$, we obtain

$$-\frac{z}{(z;q)_2}\sum_{r=0}^{N-1}\sum_{\ell=0}^{L}z\mathcal{K}_{N-1,L;r,\ell}(zq,wq^{-2};q)\Psi_{r,\ell}(zq,wq^{-2};q).$$

Applying (3.22a) with the parameter substitution $(z,w)\mapsto(zq,wq^{-2})$, we get

$$-\frac{z^2}{(z;q)_2}\Psi_{N-1,L}(zq,wq^{-2};q). \tag{3.24}$$

On the other hand, the right-hand side

$$\Psi_{N,L}(Uz,Vw;z,w;q)$$

has the U-coefficient (3.24), as follows from the second term in (3.21).

The UV-coefficient on the left is

$$\frac{zw^2}{(w;q^2)_2(zw;q)_2}\sum_{r=0}^{N}\sum_{\ell=0}^{L}\mathcal{K}_{N,L;r,\ell}(z/q,w;q)\Psi_{r-1,\ell-1}(z/q,wq^2;q).$$

Since $\Psi_{r-1,\ell-1}=0$ unless $r,\ell\geq 1$, we set

$$r=R+1,\quad \ell=S+1.$$

Then this coefficient becomes

$$\frac{zw^2}{(w;q^2)_2(zw;q)_2}\sum_{R=0}^{N-1}\sum_{S=0}^{L-1}\mathcal{K}_{N,L;R+1,S+1}(z/q,w;q)\Psi_{R,S}(z/q,wq^2;q).$$

Using the second kernel ratio in (3.23), we obtain

$$\frac{zw^2}{(w;q^2)_2(zw;q)_2}\sum_{R=0}^{N-1}\sum_{S=0}^{L-1}zw\mathcal{K}_{N-1,L-1;R,S}(z/q,wq^2;q)\Psi_{R,S}(z/q,wq^2;q).$$

By (3.22a), with the parameter change $(z,w)\mapsto(z/q,wq^2)$, we get

$$\frac{z^2w^3}{(w;q^2)_2(zw;q)_2}\Psi_{N-1,L-1}(z/q,wq^2;q).$$

This agrees with the UV-coefficient of $\Psi_{N,L}(Uz,Vw;z,w;q)$ of the right-hand side.

The U^2V-coefficient on the left is

$$\begin{aligned}&-\frac{z^4w^2}{(z;q)_2(zw;q)_2(z^2w;q^2)_2}\sum_{r=0}^{N}\sum_{\ell=0}^{L}\mathcal{K}_{N,L;r,\ell}(z/q,w;q)\Psi_{r-2,\ell-1}(zq,w;q)\\ =&-\frac{z^4w^2}{(z;q)_2(zw;q)_2(z^2w;q^2)_2}\sum_{R=0}^{N-2}\sum_{S=0}^{L-1}\mathcal{K}_{N,L;R+2,S+1}(z/q,w;q)\Psi_{R,S}(zq,w;q).\end{aligned}$$

Using the last kernel ratio in (3.23), we obtain

$$-\frac{z^4w^2}{(z;q)_2(zw;q)_2(z^2w;q^2)_2}\sum_{R=0}^{N-2}\sum_{S=0}^{L-1}z^2w\mathcal{K}_{N-2,L-1;R,S}(zq,w;q)\Psi_{R,S}(zq,w;q).$$

Applying (3.22a) with the parameter substitution $(z,w)\mapsto(zq,w)$, we get

$$-\frac{z^6w^3}{(z;q)_2(zw;q)_2(z^2w;q^2)_2}\Psi_{N-2,L-1}(zq,w;q),$$

which agrees with the U^2V-coefficient on the right-hand side. The identities also hold when one of these sums is empty, by the zero convention. This completes the proof. □

Write

$$r^2-2r\ell+2\ell^2=(r-\ell)^2+\ell^2.$$

This form motivates the shift coordinates $u=r-\ell$ and $v=\ell$. Put $a=z^2w$. For integers u,v, define the shifted functions

$$\Psi_{N,L;u,v}(z,w;q):=\frac{\Psi_{N-u-v,L-v}(zq^{2u},wq^{-2u+2v};q)}{(zq;q)_{2u}(zwq;q)_{2v}(wq^2;q^2)_{v-u}(z^2wq^2;q^2)_{u+v}}. \tag{3.25}$$

For reference, expanding (3.25) gives

$$\begin{aligned}\Psi_{N,L;u,v}(z,w;q)=&\sum_{h=0}^{\min\{\lfloor(N-u-v)/2\rfloor,L-v\}}\frac{a^hq^{2h^2+2h(u+v)}(zwq^{2L+1};q)_{N-u-v-2h}}{(q^2;q^2)_h(aq^2;q^2)_{u+v+h}(q;q)_{N-u-v-2h}}\\ &\times\frac{1}{(zq;q)_{N+u-v}(zwq;q)_{N-u+v}(q^2;q^2)_{L-v-h}(wq^2;q^2)_{L-u-h}}.\end{aligned} \tag{3.26}$$

Define the corresponding shifted deformation by

$$\Psi_{N,L;u,v}(U,V;z,w;q):=\frac{\Psi_{N-u-v,L-v}(U,V;zq^{2u},wq^{-2u+2v};q)}{D_{u,v}(z,w)}, \tag{3.27}$$

where $D_{u,v}(z,w):=(zq;q)_{2u}(zwq;q)_{2v}(wq^2;q^2)_{v-u}(z^2wq^2;q^2)_{u+v}$. Both shifted functions are zero if $N-u-v<0$ or $L-v<0$.

We define B_2 Bailey pairs on the cone

$$\mathscr{C}:=\{(u,v)\in\mathbb{Z}^2:v\geq0,\ u+v\geq0\}.$$

For fixed N,L, the nonzero terms must satisfy $u+v\leq N$ and $v\leq L$, so the pair relation is a finite sum.

Definition 3.6. Let $\{\alpha_{u,v}\}_{u,v\in\mathbb{Z}}$ be supported on $\mathscr{C}$, and let $\{\beta_{N,L}\}_{N,L\geq 0}$ satisfy

$$\beta_{N,L} = \sum_{u,v\in\mathbb{Z}} \Psi_{N,L;u,v}(z,w;q)\alpha_{u,v}.$$

Then (α,β) is called a (symmetrized) B_2 Bailey pair relative to (z,w).

The shifted chain and lattice identities below provide the two iteration rules used in Theorems 1.2 and 1.4. The chain contributes the factor $z^{u+v}w^v q^{u^2+v^2}$, while the lattice updates the deformation parameters and produces the linear terms in the multisum exponents.

Theorem 3.7. *For nonnegative integers N, L, $(u,v)\in\mathscr{C}$, and arbitrary parameters U, V,*

$$\sum_{r=0}^{N}\sum_{\ell=0}^{L} \mathcal{K}_{N,L;r,\ell}(z,w;q)\Psi_{r,\ell;u,v}(1,1;z,w;q) = z^{u+v}w^v q^{u^2+v^2}\Psi_{N,L;u,v}(1,1;z,w;q) \tag{3.28a}$$

and

$$\begin{aligned}\sum_{r=0}^{N}\sum_{\ell=0}^{L} &\mathcal{K}_{N,L;r,\ell}(z/q,w;q)\Psi_{r,\ell;u,v}(U,V;z,w;q)\\ &= z^{u+v}w^v q^{u^2+v^2-u-v}\Psi_{N,L;u,v}\big(Uzq^{2u}, Vwq^{-2u+2v};z,w;q\big).\end{aligned} \tag{3.28b}$$

Proof. If $N < u+v$ or $L < v$, both sides of each identity vanish. Otherwise, write

$$r = R+u+v = R+A, \quad \ell = S+v = S+B.$$

First compute the quadratic form:

$$\begin{aligned} r^2 - 2r\ell + 2\ell^2 &= (R+A)^2 - 2(R+A)(S+B) + 2(S+B)^2\\ &= R^2 - 2RS + 2S^2 + (2A-2B)R + (-2A+4B)S\\ &\quad + A^2 - 2AB + 2B^2.\end{aligned}$$

Substituting $A = u+v, \quad B = v$, we obtain

$$2A-2B = 2u, \quad -2A+4B = -2u+2v, \quad A^2 - 2AB + 2B^2 = u^2+v^2.$$

Therefore,

$$\mathcal{K}_{N,L;R+u+v,S+v}(z,w;q) = z^{u+v}w^v q^{u^2+v^2}\mathcal{K}_{N-u-v,L-v;R,S}(zq^{2u}, wq^{-2u+2v};q).$$

Similarly, if the kernel contains z/q, then there is an additional factor

$$(z/q)^{R+u+v} = z^{R+u+v}q^{-R-u-v}.$$

The factor q^{-R} changes the first parameter of the shifted kernel from zq^{2u} to zq^{2u-1}. Thus,

$$\mathcal{K}_{N,L;R+u+v,S+v}(z/q,w;q) = z^{u+v}w^v q^{u^2+v^2-u-v}\mathcal{K}_{N-u-v,L-v;R,S}(zq^{2u-1}, wq^{-2u+2v};q). \tag{3.29}$$

Put

$$Z = zq^{2u}, \qquad T = wq^{-2u+2v}.$$

By definition,

$$\Psi_{r,\ell;u,v}(1,1;z,w;q) = \frac{\Psi_{r-u-v,\ell-v}(1,1;Z,T;q)}{D_{u,v}(z,w)}.$$

Hence the left-hand side of (3.28a) equals

$$\frac{1}{D_{u,v}(z,w)}\sum_{R=0}^{N-u-v}\sum_{S=0}^{L-v}\mathcal{K}_{N,L;R+u+v,S+v}(z,w;q)\Psi_{R,S}(1,1;Z,T;q).$$

This is

$$z^{u+v}w^v q^{u^2+v^2}\Psi_{N,L;u,v}(1,1;z,w;q).$$

This proves (3.28a).

The left-hand side of (3.28b) equals

$$\begin{aligned}&\frac{1}{D_{u,v}(z,w)}\sum_{r=0}^{N}\sum_{\ell=0}^{L}\mathcal{K}_{N,L;r,\ell}(z/q,w;q)\Psi_{r-u-v,\ell-v}(U,V;Z,T;q)\\ =&\frac{1}{D_{u,v}(z,w)}\sum_{R=0}^{N-u-v}\sum_{S=0}^{L-v}\mathcal{K}_{N,L;R+u+v,S+v}(z/q,w;q)\Psi_{R,S}(U,V;Z,T;q).\end{aligned}$$

Using the relation (3.29), we obtain

$$\frac{z^{u+v}w^v q^{u^2+v^2-u-v}}{D_{u,v}(z,w)}\sum_{R=0}^{N-u-v}\sum_{S=0}^{L-v}\mathcal{K}_{N-u-v,L-v;R,S}(Z/q,T;q)\Psi_{R,S}(U,V;Z,T;q).$$

By the unshifted lattice identity (3.22b), the double sum equals

$$\Psi_{N-u-v,L-v}(UZ,VT;Z,T;q).$$

That is,

$$\Psi_{N-u-v,L-v}(Uzq^{2u},Vwq^{-2u+2v};zq^{2u},wq^{-2u+2v};q).$$

Dividing by $D_{u,v}(z,w)$ and using (3.27) proves (3.28b). □

The following transformation iterates a B_2 Bailey pair while preserving parameters z,w. This is the basic form of our B_2 Bailey lemma.

Lemma 3.8. *Let (α,β) be a symmetrized B_2 Bailey pair relative to (z,w). Define*

$$\alpha'_{u,v}=z^{u+v}w^v q^{u^2+v^2}\alpha_{u,v},$$

and

$$\beta'_{N,L}=\sum_{r=0}^{N}\sum_{\ell=0}^{L}\mathcal{K}_{N,L;r,\ell}(z,w;q)\beta_{r,\ell}.$$

Then (α',β') is again a (symmetrized) B_2 Bailey pair relative to (z,w).

Proof. Substitute the defining relation for $\beta_{r,\ell}$ into the formula for $\beta'_{N,L}$. All sums are finite, so their order may be interchanged. Applying (3.28a) gives

$$\begin{aligned}\beta'_{N,L}&=\sum_{(u,v)\in\mathscr{C}}\alpha_{u,v}\sum_{r=0}^{N}\sum_{\ell=0}^{L}\mathcal{K}_{N,L;r,\ell}(z,w;q)\Psi_{r,\ell;u,v}(z,w;q)\\ &=\sum_{(u,v)\in\mathscr{C}}\Psi_{N,L;u,v}(z,w;q)z^{u+v}w^v q^{u^2+v^2}\alpha_{u,v}=\sum_{(u,v)\in\mathscr{C}}\Psi_{N,L;u,v}(z,w;q)\alpha'_{u,v}.\end{aligned}$$

This is the defining relation for the transformed pair. □

4 Two B_2 Bailey pairs and a character polynomial

In this section, we construct two B_2 Bailey pairs in the sense of Definition 3.6. Their nonzero coefficients are parametrized by one and two nonnegative integers, respectively. We also identify the spinor-character polynomial used to derive the identities in the next section.

Throughout this section, put $a = z^2w$, and for each integer $u \geq 0$ define

$$C_u(a;q) := (-1)^u a^u q^{u(3u-1)} \frac{1-aq^{4u}}{1-a} \frac{(a;q^2)_u}{(q^2;q^2)_u}. \tag{4.1}$$

4.1 A diagonal Bailey pair

Theorem 4.1 (A diagonal Bailey pair)**.** *The sequences*

$$\alpha_{u,v} = \begin{cases} C_u(a;q), & u = v \geq 0, \\ 0, & \text{otherwise}, \end{cases} \qquad \beta_{N,L} = \frac{(zwq^{2L+1};q)_N}{(q, zq, zwq;q)_N (q^2, wq^2;q^2)_L} := \mathcal{A}_{N,L}(z,w;q)$$

form a B_2 Bailey pair relative to (z,w). Equivalently,

$$\sum_{j\geq 0} C_j(a;q) \Psi_{N,L;j,j}(z,w;q) = \mathcal{A}_{N,L}(z,w;q). \tag{4.2}$$

Proof. For $u = v = j$, (3.26) becomes

$$\begin{aligned} \Psi_{N,L;j,j}(z,w;q) = \sum_{h\geq 0} & \frac{a^h q^{2h^2+4jh} (zwq^{2L+1};q)_{N-2j-2h}}{(q^2;q^2)_h (aq^2;q^2)_{2j+h} (q;q)_{N-2j-2h}} \\ & \times \frac{1}{(zq, zwq;q)_N (q^2;q^2)_{L-j-h} (wq^2;q^2)_{L-j-h}}. \end{aligned} \tag{4.3}$$

Insert (4.3) into the left side of (4.2). Set $k = j + h$. Then the left side equals

$$\frac{1}{(zq, zwq;q)_N} \sum_{k\geq 0} \frac{a^k q^{2k^2} (zwq^{2L+1};q)_{N-2k}}{(q;q)_{N-2k} (q^2;q^2)_{L-k} (wq^2;q^2)_{L-k}} I_k, \tag{4.4}$$

where

$$I_k := \sum_{j=0}^{k} \frac{(-1)^j q^{j(j-1)} (1-aq^{4j}) (a;q^2)_j}{(1-a)(q^2;q^2)_j (q^2;q^2)_{k-j} (aq^2;q^2)_{k+j}} = \frac{T_k}{(q^2;q^2)_k (aq^2;q^2)_k},$$

and

$$T_k = \sum_{j=0}^{k} \frac{1-aq^{4j}}{1-a} \frac{(a, q^{-2k};q^2)_j}{(q^2, aq^{2k+2};q^2)_j} q^{2kj}.$$

It is the terminating very-well-poised series

$$T_k = {}_4\phi_3 \left[\begin{matrix} a, & q^2\sqrt{a}, & -q^2\sqrt{a}, & q^{-2k} \\ \sqrt{a}, & -\sqrt{a}, & aq^{2k+2} \end{matrix} ; q^2, q^{2k} \right].$$

This is obtained from Jackson's terminating very-well-poised ${}_6\phi_5$ summation (2.2) by choosing $bc = aq^2$. With this specialization, the b- and c-dependent factors cancel pairwise in the summand, while the product side contains $(1;q^2)_k$. Therefore

$$T_k = \begin{cases} 1, & k = 0, \\ 0, & k > 0. \end{cases}$$

Consequently $I_k = \delta_{k,0}$. In (4.4), only the $k = 0$ term remains, giving exactly $\mathcal{A}_{N,L}(z,w;q)$. □

4.2 A two-index Bailey pair

We first shift the identity of Theorem 4.1. Define

$$\begin{aligned}\mathcal{B}_{N,L;u,v}(z,w) &:= \frac{\mathcal{A}_{N-u-v,L-v}(zq^{2u},wq^{-2u+2v})}{D_{u,v}(z,w)}\\ &= \frac{(zwq^{2L+1};q)_{N-u-v}}{(q;q)_{N-u-v}(zq;q)_{N+u-v}(zwq;q)_{N-u+v}}\,\frac{1}{(q^2;q^2)_{L-v}(wq^2;q^2)_{L-u}(aq^2;q^2)_{u+v}}.\end{aligned} \tag{4.5}$$

Proposition 4.2. *Let N, L be nonnegative integers and let $u, v \in \mathbb{Z}$ satisfy $N-u-v \geq 0$ and $L-v \geq 0$. Then*

$$\mathcal{B}_{N,L;u,v}(z,w) = \sum_{j\geq 0} C_j(aq^{2u+2v};q)\Psi_{N,L;u+j,v+j}(z,w). \tag{4.6}$$

Proof. Put

$$Z = zq^{2u}, \qquad T = wq^{-2u+2v}.$$

Then $Z^2T = aq^{2u+2v}$. Apply (4.2) with (N,L,z,w) replaced by $(N-u-v, L-v, Z, T)$:

$$\mathcal{A}_{N-u-v,L-v}(Z,T) = \sum_{j\geq 0} C_j(aq^{2u+2v};q)\Psi_{N-u-v,L-v;j,j}(Z,T).$$

Dividing both sides by $D_{u,v}(z,w)$ gives

$$\mathcal{B}_{N,L;u,v}(z,w) = \sum_{j\geq 0} C_j(aq^{2u+2v};q)\frac{\Psi_{N-u-v,L-v;j,j}(Z,T)}{D_{u,v}(z,w)}.$$

The factors satisfy

$$\begin{aligned}&D_{j,j}(Z,T) = (zq^{2u+1};q)_{2j}(zwq^{2v+1};q)_{2j}(aq^{2u+2v+2};q^2)_{2j},\\ &D_{u,v}(z,w)D_{j,j}(Z,T) = D_{u+j,v+j}(z,w).\end{aligned}$$

We arrive at

$$\begin{aligned}\frac{\Psi_{N-u-v,L-v;j,j}(Z,T)}{D_{u,v}(z,w)} &= \frac{\Psi_{N-(u+j)-(v+j),L-(v+j)}(zq^{2(u+j)},wq^{-2(u+j)+2(v+j)})}{D_{j,j}(Z,T)D_{u,v}(z,w)}\\ &= \Psi_{N,L;u+j,v+j}(z,w).\end{aligned}$$

This establishes the claimed identity. □

Now set

$$u = -H, \qquad v = H, \qquad H \geq 0.$$

Then $u+v=0$ and (4.5) gives

$$\mathcal{B}_{N,L;-H,H}(z,w) = \frac{(zwq^{2L+1};q)_N}{(q;q)_N(zq;q)_{N-2H}(zwq;q)_{N+2H}(q^2;q^2)_{L-H}(wq^2;q^2)_{L+H}}. \tag{4.7}$$

We next sum the identity over $(u,v) = (-H,H)$ and hence find a second explicit Bailey pair.

Theorem 4.3 (A two-index Bailey pair)**.** *For $N, L \geq 0$,*

$$\sum_{H\geq 0} C_H(w;q)\mathcal{B}_{N,L;-H,H}(z,w) = \mathcal{R}_{N,L}(z,w), \tag{4.8}$$

where $\mathcal{R}_{N,L}$ is given by (1.5). The resulting B_2 Bailey pair is

$$\alpha_{u,v} = \begin{cases} C_j(a;q)C_H(w;q), & u = j-H,\ v = j+H, \quad j,H \in \mathbb{Z}_{\geq 0}, \\ 0, & \text{otherwise}, \end{cases} \qquad \beta_{N,L} = \mathcal{R}_{N,L}(z,w).$$

Proof. Using (4.7) and dividing the left-hand side of (4.8) by

$$\frac{(zwq^{2L+1};q)_N}{(q,zq,zwq;q)_N(q^2,wq^2;q^2)_L}$$

leaves the sum

$$\begin{aligned} S_{N,L} &= \sum_{H\geq 0} C_H(w;q)\frac{(zq;q)_N(zwq;q)_N(q^2;q^2)_L(wq^2;q^2)_L}{(zq;q)_{N-2H}(zwq;q)_{N+2H}(q^2;q^2)_{L-H}(wq^2;q^2)_{L+H}} \\ &= {}_6W_5\left(w;q^{-N}/z,q^{1-N}/z,q^{-2L};q^2,aq^{2N+2L+1}\right), \end{aligned}$$

where we have used

$$\begin{aligned} &\frac{(zq;q)_N}{(zq;q)_{N-2H}} = z^{2H}q^{2HN+H-2H^2}(q^{-N}/z,q^{1-N}/z;q^2)_H, \\ &\frac{(q^2;q^2)_L}{(q^2;q^2)_{L-H}} = (-1)^H q^{2LH-H(H-1)}(q^{-2L};q^2)_H, \\ &\frac{(zwq;q)_N}{(zwq;q)_{N+2H}} = \frac{1}{(zwq^{N+1},zwq^{N+2};q^2)_H}, \frac{(wq^2;q^2)_L}{(wq^2;q^2)_{L+H}} = \frac{1}{(wq^{2L+2};q^2)_H}. \end{aligned}$$

The terminating very-well-poised ${}_6\phi_5$ summation (2.2) gives

$$S_{N,L} = \frac{(wq^2,aq^{2N+1};q^2)_L}{(zwq^{N+2},zwq^{N+1};q^2)_L}.$$

After substituting this expression into (4.8), the factor $(wq^2;q^2)_L$ cancels, and

$$(zwq^{N+2},zwq^{N+1};q^2)_L = (zwq^{N+1};q)_{2L},$$

which yields the product in (1.5).

To obtain the pair relation, set $u = -H$ and $v = H$ in (4.6). This gives

$$\mathcal{B}_{N,L;-H,H}(z,w) = \sum_{j\geq 0} C_j(a;q)\Psi_{N,L;j-H,j+H}(z,w).$$

Therefore (4.8) becomes

$$\mathcal{R}_{N,L}(z,w) = \sum_{j,H\geq 0} C_j(a;q)C_H(w;q)\Psi_{N,L;j-H,j+H}(z,w). \tag{4.9}$$

This proves the stated pair relation. □

There is also a direct root-system description of the two indices. Since $r = u+v$ and $\ell = v$, the vector associated with $(u,v) = (j-H, j+H)$ is

$$ve_1 + ue_2 = j(e_1+e_2) + H(e_1-e_2).$$

Thus j, H parametrize the two positive long-root directions of B_2.

4.3 The spinor-character polynomial

Let h_s denote the complete homogeneous symmetric polynomial of degree s, with $h_0 = 1$, and put $W = Z^2T$. For every integer $s \geq 0$, define

$$E_s(Z,T) := h_s(1, Z, ZT, Z^2T).$$

Proposition 4.4 (The spinor-character polynomial)**.** *For every nonnegative integer s, we have*

$$E_s(Z,T) = \sum_{r=0}^{s} Z^r(1 + T + \cdots + T^r)(1 + W + \cdots + W^{s-r}), \tag{4.10}$$

$$\sum_{s\geq 0} E_s(Z,T)x^s = \frac{1}{(1-x)(1-Zx)(1-ZTx)(1-Wx)}. \tag{4.11}$$

Equivalently,

$$E_s(Z,T) = \frac{1}{(1-Z)(1-ZT)}\left(\frac{1-W^{s+2}}{1-W} - Z^{s+1}\frac{1-T^{s+2}}{1-T}\right). \tag{4.12}$$

For the simple roots $\alpha_1 = e_2$, $\alpha_2 = e_1 - e_2$ of B_2, let $\omega_{\mathrm{sp}} = (e_1 + e_2)/2$. Then

$$E_s(Z,T) = e^{-s\omega_{\mathrm{sp}}} \operatorname{ch} V_{B_2}(s\omega_{\mathrm{sp}})\big|_{e^{-\alpha_1}=Z,\ e^{-\alpha_2}=T}. \tag{4.13}$$

Proof. Splitting the four variables into $(1, W)$ and (Z, ZT) gives

$$h_s(1, W, Z, ZT) = \sum_{r=0}^{s} h_{s-r}(1, W)h_r(Z, ZT).$$

Since $h_r(Z, ZT) = Z^r(1 + T + \cdots + T^r)$, this proves (4.10); (4.11) is the defining generating function of h_s. For generic Z, T, the four poles in (4.11) are distinct, and

$$\frac{1}{(1-x)(1-Zx)(1-ZTx)(1-Wx)} = \frac{A}{1-x} + \frac{B}{1-Zx} + \frac{C}{1-ZTx} + \frac{D}{1-Wx},$$

where multiplication by the corresponding linear factor and evaluation at its zero gives

$$A = \frac{1}{(1-Z)(1-ZT)(1-W)}, \quad B = -\frac{Z}{(1-Z)(1-T)(1-ZT)},$$
$$C = \frac{ZT^2}{(1-Z)(1-T)(1-ZT)}, \quad D = -\frac{W^2}{(1-Z)(1-ZT)(1-W)}.$$

Here the expression for D uses $W = Z^2T$. Extracting the coefficient of x^s yields $E_s = A + BZ^s + C(ZT)^s + DW^s$, which simplifies to (4.12). Since E_s is a polynomial, the apparent singularities in that expression are removable after cancellation.

We next prove (4.13). The fundamental weights for our choice of simple roots are

$$\omega_1 = \frac{e_1 + e_2}{2} = \omega_{\mathrm{sp}}, \qquad \omega_2 = e_1.$$

The spinor representation $S = V_{B_2}(\omega_{\mathrm{sp}})$ is four-dimensional, with weights $\frac{1}{2}(\pm e_1 \pm e_2)$, each of multiplicity one [13, 23]. Its normalized character is therefore

$$e^{-\omega_{\mathrm{sp}}} \operatorname{ch} S = 1 + e^{-\alpha_1} + e^{-(\alpha_1+\alpha_2)} + e^{-(2\alpha_1+\alpha_2)}.$$

Under the specialization in (4.13), these four terms become $1, Z, ZT, W$.

For completeness, $\mathrm{Sym}^s S$ is irreducible in this case. The s-th power of a highest-weight vector in S is a nonzero highest-weight vector of weight $s\omega_{\mathrm{sp}}$, so $V_{B_2}(s\omega_{\mathrm{sp}})$ occurs in $\mathrm{Sym}^s S$. The positive roots are $e_2, e_1 - e_2, e_1, e_1 + e_2$, whose half-sum is $\rho_B = (3e_1 + e_2)/2$. The Weyl dimension formula gives

$$\dim V_{B_2}(s\omega_{\mathrm{sp}}) = \frac{(s+1)(s+2)(s+3)}{6} = \dim \mathrm{Sym}^s S.$$

Complete reducibility now implies $V_{B_2}(s\omega_{\mathrm{sp}}) \cong \mathrm{Sym}^s S$. Taking the character of this symmetric power proves (4.13). □

The expansion (4.10) also records the restriction to the long-root subalgebra. Indeed, the two positive long roots are $\beta_+ = e_1 + e_2$ and $\beta_- = e_1 - e_2$, and they are orthogonal. The root spaces for $\pm\beta_+$ and $\pm\beta_-$, together with their coroots, form a subalgebra of type $D_2 = A_1 \oplus A_1$. Write U_m for the irreducible A_1-module of highest weight m, and use $\boxtimes$ for the external tensor product. The two pairs of weights $\pm\beta_+/2$ and $\pm\beta_-/2$ give

$$S{\downarrow_{D_2}} \cong (U_1 \boxtimes U_0) \oplus (U_0 \boxtimes U_1).$$

The standard decomposition [15]

$$\mathrm{Sym}^s(A \oplus B) \cong \bigoplus_{r=0}^{s} \mathrm{Sym}^{s-r} A \otimes \mathrm{Sym}^r B,$$

together with $\mathrm{Sym}^m U_1 \cong U_m$, yields

$$V_{B_2}(s\omega_{\mathrm{sp}}){\downarrow_{D_2}} \cong \bigoplus_{r=0}^{s} U_{s-r} \boxtimes U_r. \tag{4.14}$$

The two normalized pairs of weights are $(1, W)$ and (Z, ZT), so (4.10) is precisely the character form of (4.14). For example,

$$E_0 = 1, \qquad E_1 = 1 + Z + ZT + Z^2T, \qquad E_s(1,1) = \binom{s+3}{3}.$$

We shall also use the polynomial

$$R_s(Z,T) := (1-W)E_s(Z,T) = \sum_{r=0}^{s} Z^r(1 + T + \cdots + T^r)(1 - W^{s+1-r}). \tag{4.15}$$

In the proof of Theorem 1.4, the factor $1 - W$ in $C_j(q^2; q)$ combines with E_s to give R_s.

5 Rogers–Ramanujan-type identities

We now iterate the chain and lattice transformations from the two Bailey pairs of Section 4. The resulting limits are evaluated using the polynomial E_s from Proposition 4.4.

5.1 The common limit and the theta identities

Put

$$\mathscr{D}(z, w; q) := (q, zq, zwq; q)_\infty (q^2, wq^2, z^2wq^2; q^2)_\infty. \tag{5.1}$$

Lemma 5.1. *Set $Z = zq^{2u}$, $T = wq^{-2u+2v}$, and $W = Z^2T$. Then, for every integer $s \geq 0$,*

$$\lim_{N,L\to\infty} \Psi_{N,L;u,v}(Z^s, T^s; z, w; q) = \frac{E_s(Z,T)}{\mathscr{D}(z,w;q)}. \tag{5.2}$$

Proof. Taking $N, L \to \infty$ in (3.3) gives, with $a = z^2w$,

$$\frac{1}{(q, zq, zwq; q)_\infty (q^2, wq^2; q^2)_\infty} \sum_{j\geq 0} \frac{a^j q^{2j^2}}{(q^2;q^2)_j (aq^2;q^2)_j},$$

where the series equals $(aq^2;q^2)_\infty^{-1}$. To see this directly, let the upper index tend to infinity in the scalar q-Chu–Vandermonde identity used in the proof of Lemma 3.3. Hence

$$\lim_{N,L\to\infty} \Psi_{N,L}(z,w;q) = \mathscr{D}(z,w;q)^{-1}.$$

Apply the same limit to the four terms in (3.21). Dividing by the unshifted limit at (Z,T), the four contributions are

$$\frac{1}{(1-Z)(1-ZT)(1-W)}, \quad -\frac{UZ}{(1-Z)(1-ZT)(1-T)},$$
$$\frac{UVZT^2}{(1-Z)(1-ZT)(1-T)}, \quad -\frac{U^2VW^2}{(1-Z)(1-ZT)(1-W)}.$$

Their sum is

$$\frac{1}{(1-Z)(1-ZT)}\left(\frac{1-U^2VW^2}{1-W} - UZ\frac{1-VT^2}{1-T}\right).$$

For $U = Z^s$, $V = T^s$, this is $E_s(Z,T)$ by (4.12). Finally, the product-shift relations give

$$D_{u,v}(z,w)\mathscr{D}(Z,T;q) = \mathscr{D}(z,w;q).$$

Dividing by $D_{u,v}$ proves (5.2). □

We use the Jacobi triple product in the form

$$\sum_{n\in\mathbb{Z}} z^n q^{n^2} = (q^2;q^2)_\infty (-zq, -q/z; q^2)_\infty. \tag{5.3}$$

For comparison, we recall the $D_n^{(1)}$ Macdonald identity [29], in the form given in [22, Appendix A]:

$$\sum_{\substack{r\in\mathbb{Z}^n \\ |r| \text{ even}}} \Delta_D(xq^r) \prod_{i=1}^n q^{2(n-1)\binom{r_i}{2}+(i-1)r_i} x_i^{2(n-1)r_i} = (q;q)_\infty^n \prod_{i<j} \theta(x_i/x_j, x_ix_j; q), \tag{5.4}$$

where $n \geq 2$, $|r| = r_1 + \cdots + r_n$, and $\Delta_D(x) = \prod_{i<j}(1-x_i/x_j)(1-x_ix_j)$. When $n = 2$, the condition $r_1 + r_2$ even is equivalent to $r_1 = m+d$, $r_2 = m-d$ for $m, d \in \mathbb{Z}$. Thus the left-hand side factors as

$$\left[\sum_{d\in\mathbb{Z}} q^{2d^2-d}(x_1/x_2)^{2d}(1-(x_1/x_2)q^{2d})\right]\left[\sum_{m\in\mathbb{Z}} q^{2m^2-m}(x_1x_2)^{2m}(1-x_1x_2q^{2m})\right]. \tag{5.5}$$

For example, writing $y = x_1/x_2$, the first factor is

$$\sum_{d\in\mathbb{Z}} q^{2d^2-d}y^{2d}(1-yq^{2d}) = \sum_{n\in\mathbb{Z}} (-1)^n y^n q^{n(n-1)/2} = (q;q)_\infty \theta(y;q),$$

where the two terms correspond to even and odd n, respectively. The other factor has the same form with $y = x_1x_2$. This is the $D_2 = A_1 \oplus A_1$ factorization used below.

5.2 Proof of Theorem 1.2

Proof. Fix $k \geq 1$ and $0 \leq s \leq k$. Write $a = z^2 w$, and put $\lambda_0 = N$, $\mu_0 = L$. Apply $k - s$ iterations of (3.28a) to the diagonal pair of Theorem 4.1. Each step multiplies its j-th coefficient by $a^j q^{2j^2}$. Thus

$$\begin{aligned}\sum_{\substack{N=\lambda_0\geq\lambda_1\geq\cdots\geq\lambda_{k-s}\geq 0\\ L=\mu_0\geq\mu_1\geq\cdots\geq\mu_{k-s}\geq 0}} \prod_{i=1}^{k-s} \frac{z^{\lambda_i} w^{\mu_i} q^{\lambda_i^2-2\lambda_i\mu_i+2\mu_i^2}}{(q;q)_{\lambda_{i-1}-\lambda_i}(q^2;q^2)_{\mu_{i-1}-\mu_i}} \mathcal{A}_{\lambda_{k-s},\mu_{k-s}}(z,w) \\ = \sum_{j\geq 0} C_j(a;q) a^{(k-s)j} q^{2(k-s)j^2} \Psi_{N,L;j,j}(1,1;z,w;q).\end{aligned} \tag{5.6}$$

Now apply s lattice transformations (3.28b). Together, these steps contribute the factor $a^{sj} q^{2sj^2-2sj}$, while the deformation parameters become $((zq^{2j})^s, w^s)$. Consequently,

$$\begin{aligned}\sum_{\substack{N=\lambda_0\geq\lambda_1\geq\cdots\geq\lambda_k\geq 0\\ L=\mu_0\geq\mu_1\geq\cdots\geq\mu_k\geq 0}} \prod_{i=1}^{k} \frac{z^{\lambda_i} w^{\mu_i} q^{\lambda_i^2-2\lambda_i\mu_i+2\mu_i^2-\chi(i\leq s)\lambda_i}}{(q;q)_{\lambda_{i-1}-\lambda_i}(q^2;q^2)_{\mu_{i-1}-\mu_i}} \mathcal{A}_{\lambda_k,\mu_k}(z,w) \\ = \sum_{j\geq 0} C_j(a;q) a^{kj} q^{2kj^2-2sj} \Psi_{N,L;j,j}((zq^{2j})^s, w^s; z, w; q).\end{aligned} \tag{5.7}$$

Multiplying the left-hand side of (5.7) by $(q;q)_\infty (q^2;q^2)_\infty$ and taking $N, L \to \infty$ gives

$$\mathcal{F}_{k,s}(z,w) := \sum_{\substack{\lambda_1\geq\cdots\geq\lambda_k\geq 0\\ \mu_1\geq\cdots\geq\mu_k\geq 0}} \frac{z^{\sum_{i=1}^k \lambda_i} w^{\sum_{i=1}^k \mu_i} q^{\sum_{i=1}^k (\lambda_i^2-2\lambda_i\mu_i+2\mu_i^2)-\sum_{i=1}^s \lambda_i}}{\prod_{i=1}^{k-1}(q;q)_{\lambda_i-\lambda_{i+1}}(q^2;q^2)_{\mu_i-\mu_{i+1}}} \mathcal{A}_{\lambda_k,\mu_k}(z,w). \tag{5.8}$$

Set $a = q^2$, or equivalently $z^2 w = q^2$. By Lemma 5.1, taking the same limit in (5.7) gives

$$\mathcal{F}_{k,s}(z,w) = \frac{(q;q)_\infty (q^2;q^2)_\infty}{\mathscr{D}(z,w;q)} \sum_{j\geq 0} C_j(q^2;q) q^{2kj+2kj^2-2sj} E_s(zq^{2j}, w). \tag{5.9}$$

Here $Z = zq^{2j}$, $T = w$, and $W = q^{4j+2}$. Since

$$C_j(q^2;q) = (-1)^j q^{3j^2+j} \frac{1-q^{4j+2}}{1-q^2},$$

(4.15) transforms the sum in (5.9) into

$$\begin{aligned}&\frac{1}{1-q^2} \sum_{r=0}^{s} z^r (1 + w + \cdots + w^r) \\ &\quad \times \sum_{j\geq 0} (-1)^j q^{(2k+3)j(j+1)-2(s+1-r)j} \big(1 - q^{2(s+1-r)(2j+1)}\big).\end{aligned} \tag{5.10}$$

Put $M = 2k + 3$ and $A = 2(s + 1 - r)$. The inner sum in (5.10) is

$$\sum_{n\in\mathbb{Z}} (-1)^n q^{Mn(n+1)-An}.$$

Indeed, its terms with $n = j$ and $n = -j - 1$ give the two terms in the corresponding unilateral summand. The Jacobi triple product, with base q^{2M}, evaluates this bilateral series as $(q^A, q^{2M-A}, q^{2M}; q^{2M})_\infty = \Theta^k_{s-r}$, where

$$\Theta^k_s := (q^{2s+2}, q^{4k+4-2s}, q^{4k+6}; q^{4k+6})_\infty.$$

We therefore obtain

$$\mathcal{F}_{k,s}(z,w)=\frac{\displaystyle\sum_{r=0}^{s} z^r(1+w+\cdots+w^r)\Theta_{s-r}^{k}}{(zq,zwq;q)_\infty(wq^2;q^2)_\infty(q^2;q^2)_\infty},\qquad z^2w=q^2. \tag{5.11}$$

Put

$$G_s:=\sum_{r=0}^{s} z^r(1+w+\cdots+w^r)\Theta_{s-r}^{k}=\sum_{r=0}^{s} c_r\Theta_{s-r}^{k}, \tag{5.12}$$

where $c_r=z^r(1+w+\cdots+w^r)$. Its generating function is

$$C(x):=\sum_{r\geq 0} c_r x^r=\frac{1}{(1-zx)(1-zwx)}.$$

For the generating-function argument, (5.12) defines G_s formally for all $s\geq 0$; its identification with the numerator in (5.11) is used only for $0\leq s\leq k$. Consider the formal generating functions

$$G(x):=\sum_{s\geq 0} G_s x^s,\qquad \Theta^k(x):=\sum_{s\geq 0}\Theta_s^k x^s.$$

The convolution (5.12) gives

$$G(x)=C(x)\Theta^k(x)=\frac{\Theta^k(x)}{(1-zx)(1-zwx)}.$$

Hence

$$\Theta^k(x)=(1-zx)(1-zwx)G(x).$$

Comparing the coefficients of x^s, we obtain

$$\Theta_s^k=G_s-z(1+w)G_{s-1}+z^2wG_{s-2},\qquad 0\leq s\leq k, \tag{5.13}$$

where $G_{-1}=G_{-2}=0$.

Dividing (5.13) by $(zq,zwq;q)_\infty(wq^2;q^2)_\infty(q^2;q^2)_\infty$ and defining

$$\mathcal{P}_{k,s}(z,w):=\mathcal{F}_{k,s}(z,w)-z(1+w)\mathcal{F}_{k,s-1}(z,w)+z^2w\mathcal{F}_{k,s-2}(z,w),$$

with $\mathcal{F}_{k,-1}=\mathcal{F}_{k,-2}=0$, we obtain

$$\mathcal{P}_{k,s}(z,w)=\frac{(q^{2s+2},q^{4k+4-2s},q^{4k+6};q^{4k+6})_\infty}{(zq,zwq;q)_\infty(wq^2;q^2)_\infty(q^2;q^2)_\infty},\quad 0\leq s\leq k,\quad z^2w=q^2.$$

Finally, substituting the definition of $\mathcal{F}_{k,s}(z,w)$ into $\mathcal{P}_{k,s}(z,w)$ gives exactly the multisum in Theorem 1.2. This completes the proof.

□

5.3 Proof of Theorem 1.4

Proof. For the two-index pair of Theorem 4.3, put $u=j-H$, $v=j+H$. The factors contributed by the chain and lattice transformations are, respectively,

$$a^j w^H q^{2j^2+2H^2},\qquad a^j w^H q^{2j^2+2H^2-2j},\qquad a=z^2w.$$

After $k-s$ chain steps and s lattice steps, the finite identity has left-hand side

$$\sum_{\substack{N=\lambda_0\geq\lambda_1\geq\cdots\geq\lambda_k\geq 0\\ L=\mu_0\geq\mu_1\geq\cdots\geq\mu_k\geq 0}}\prod_{i=1}^{k}\frac{z^{\lambda_i}w^{\mu_i}q^{\lambda_i^2-2\lambda_i\mu_i+2\mu_i^2-\chi(i\leq s)\lambda_i}}{(q;q)_{\lambda_{i-1}-\lambda_i}(q^2;q^2)_{\mu_{i-1}-\mu_i}}\mathcal{R}_{\lambda_k,\mu_k}(z,w), \tag{5.14}$$

and right-hand side

$$\begin{aligned}\sum_{j,H\geq 0} C_j(a;q)C_H(w;q)a^{kj}w^{kH}q^{2k(j^2+H^2)-2sj}\\ \times\Psi_{N,L;j-H,j+H}((zq^{2j-2H})^s,(wq^{4H})^s;z,w;q).\end{aligned} \tag{5.15}$$

Take the limit first for generic z,w and then specialize to $w=q^2$, $z^2=1$, so that $a=q^2$. Let $\mathcal{M}_{k,s}(z;q)$ denote the multisum in (1.6). Multiplying by $(q;q)_\infty(q^2;q^2)_\infty$, taking $N,L\to\infty$, and applying Lemma 5.1 gives

$$\begin{aligned}\mathcal{M}_{k,s}(z;q)=\frac{(q;q)_\infty(q^2;q^2)_\infty}{\mathscr{D}(z,q^2;q)}\sum_{j,H\geq 0}C_j(q^2;q)C_H(q^2;q)\\ \times q^{2k(j^2+H^2)+2k(j+H)-2sj}E_s(zq^{2j-2H},q^{4H+2}).\end{aligned} \tag{5.16}$$

In this formula

$$Z=zq^{2j-2H},\qquad T=q^{4H+2},\qquad W=q^{4j+2}.$$

Using $(1-W)E_s=R_s$, we combine E_s with the factor $1-q^{4j+2}$ in $C_j(q^2;q)$. The resulting polynomial is

$$R_s(Z,T)=\sum_{r=0}^{s}\sum_{d=0}^{r}z^r q^{2d+2jr+(4d-2r)H}\bigl(1-q^{2(s+1-r)(2j+1)}\bigr). \tag{5.17}$$

Thus the sum on the right of (5.16), including its pair coefficients but not the outer product factor, becomes

$$\frac{1}{(1-q^2)^2}\sum_{r=0}^{s}z^r\,J_{k,s-r}\,H_{k,r}, \tag{5.18}$$

where

$$J_{k,s-r}:=\sum_{j\geq 0}(-1)^j q^{(2k+3)j(j+1)-2(s+1-r)j}\bigl(1-q^{2(s+1-r)(2j+1)}\bigr)=\Theta^k_{s-r}$$

by the same substitution in (5.3) as above. For the other factor, first sum over d:

$$\sum_{d=0}^{r}q^{2d+(4d-2r)H}=q^{-2rH}\frac{1-q^{2(r+1)(2H+1)}}{1-q^{4H+2}}.$$

The denominator cancels the factor $1-q^{4H+2}$ in $C_H(q^2;q)$, giving

$$H_{k,r}:=\sum_{H\geq 0}(-1)^H q^{(2k+3)H(H+1)-2(r+1)H}\bigl(1-q^{2(r+1)(2H+1)}\bigr)=\Theta^k_r.$$

These two rank-one evaluations realize the factorization illustrated by (5.5).

Finally,

$$\mathscr{D}(z,q^2;q)=(q,zq,zq^3;q)_\infty(q^2;q^2)_\infty(q^4;q^2)_\infty^2,$$

and $(1-q^2)(q^4;q^2)_\infty=(q^2;q^2)_\infty$. Combining (5.16) and (5.18) therefore yields

$$\mathcal{M}_{k,s}(z;q)=\frac{\displaystyle\sum_{r=0}^{s}z^r\Theta^k_{s-r}\Theta^k_r}{(zq,zq^3;q)_\infty(q^2;q^2)_\infty^2},$$

as required. □

6 Character specializations and CMPP partitions

We verify the normalization in Corollary 1.3. For $\ell \geq 0$, $0 \leq b \leq \ell$, and $\Lambda = (\ell - b)\Lambda_0 + b\Lambda_1$, the principal specialization is

$$e^{-\Lambda} \operatorname{ch} L(\Lambda)\big|_{e^{-\alpha_0}=e^{-\alpha_1}=p} = \frac{(p^{b+1}, p^{\ell-b+1}, p^{\ell+2}; p^{\ell+2})_\infty}{(p;p)_\infty (p;p^2)_\infty}. \tag{6.1}$$

This is the rank-one principal-specialization formula [21, 27]; see also [22, Section 4]. Taking $p = q^2$, $\ell = 2k+1$, and $b = a$, and using $(p;p^2)_\infty^{-1} = (-p;p)_\infty$, gives

$$\Xi_{k,a}(q) := (-q^2;q^2)_\infty \frac{\Theta_a^k}{(q^2;q^2)_\infty} = e^{-\Lambda} \operatorname{ch} L(\Lambda)\big|_{e^{-\alpha_0}=e^{-\alpha_1}=q^2}. \tag{6.2}$$

On the other hand, Theorem 1.2 at $z = 1, w = q^2$ gives

$$B_2^{a,k}(1,q) = \frac{\Theta_a^k}{(q,q^3;q)_\infty (q^4;q^2)_\infty (q^2;q^2)_\infty}.$$

Multiplying by $(q,q^3;q)_\infty (q^4;q^2)_\infty (-q^2;q^2)_\infty$ proves Corollary 1.3. In particular, the theta modulus $2k+3$, measured in powers of q^2, corresponds to level $2k+1$.

The second multisum admits a corresponding character specialization. For $k \geq 1$ and $0 \leq s \leq k$, let $\mathcal{M}_{k,s}(z;q)$ denote the left-hand side of (1.6). Then

$$(q,q^3;q)_\infty (-q^2;q^2)_\infty^2 \mathcal{M}_{k,s}(1;q) = \sum_{r=0}^{s} \Xi_{k,s-r}(q)\Xi_{k,r}(q). \tag{6.3}$$

The right-hand side is a sum of normalized principal specializations of $A_1^{(1)} \oplus A_1^{(1)}$ tensor-product characters. Together with Proposition 4.4, this explains the finite character polynomial and the two affine rank-one factors.

Several specializations of Theorem 1.2 reproduce extremal CMPP products. We record these comparisons using the notation of [22]. Fix nonnegative integers $k_0, \dots, k_n$. For an n-coloured partition λ, let $f_i^{(c)}$ be the multiplicity of the part i of colour c, and put

$$f_0^{(c)} = k_0 \delta_{c,1}, \qquad f_{-1}^{(c)} = k_c.$$

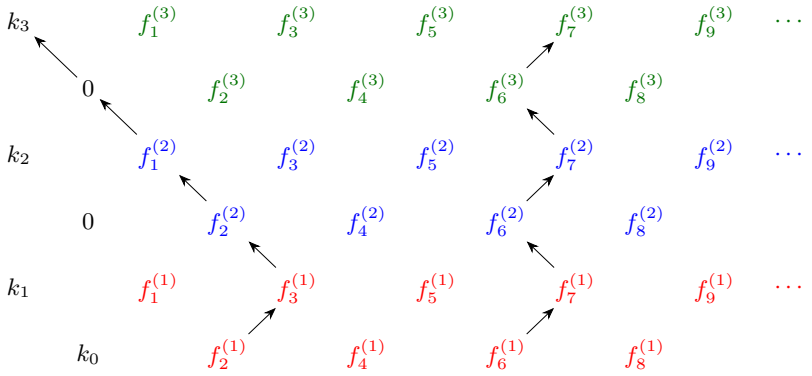
Arrange the frequencies in the alternating $2n$-row array $F_n(\lambda)$ shown below. A path $(p_1, \dots, p_{2n})$ starts in the bottom row, takes one entry from each row, and satisfies

$$p_{2c-1} \in \{f_0^{(c)}, f_2^{(c)}, f_4^{(c)}, \dots\}, \qquad p_{2c} \in \{f_{-1}^{(c)}, f_1^{(c)}, f_3^{(c)}, \dots\}.$$

The part index changes by 1 or -1 at each step, with no index smaller than -1. Following [9,22], we call λ a CMPP partition if every such path satisfies

$$\sum_{i=1}^{2n} p_i \leq k_0 + \cdots + k_n.$$

Write $\mathcal{A}_{k_0,\dots,k_n}$ for the set of these partitions. The diagram illustrates two paths for $n = 3$.



Conjecture 6.1 ([9, Conjecture 4.1]). *Let $n \geq 1$, and let $k_0, \ldots, k_n$ be nonnegative integers with sum k. Then*

$$\sum_{\lambda \in \mathcal{A}_{k_0,\ldots,k_n}} q^{|\lambda|} = \frac{(q^{2k+2n+1};q^{2k+2n+1})_\infty^n}{(q;q)_\infty^n} \prod_{i=1}^{n} \theta(q^{\kappa_i+n-i+1};q^{2k+2n+1}) \times \prod_{1\leq i<j\leq n} \theta(q^{\kappa_i-\kappa_j-i+j}, q^{\kappa_i+\kappa_j+2n-i-j+2};q^{2k+2n+1}),$$

where $\kappa_i := k_i + \cdots + k_n$ for $1 \leq i \leq n$.

For $n = 1$, the path conditions are $f_i + f_{i+1} \leq k_0 + k_1$, $f_1 \leq k_1$, which are Gordon's frequency conditions.

The two further families have parity restrictions. In $\mathcal{C}_{k_0,\ldots,k_n}$, the array has $2n+1$ colour labels and only entries with $i + c$ even are allowed; in $\mathcal{D}_{k_0,\ldots,k_n}$, there are $2n-1$ colour labels and $i + c$ is odd. The boundary data and admissible paths, including the one-row convention for $\mathcal{D}$ when $n = 1$, are those of [22, Section 4]; see also [9, 10, 34]. Both families use the path bound $k_0 + \cdots + k_n$.

Conjecture 6.2 ([22, Conjecture 4.1]). *Let $n \geq 1$, and let $k_0, \ldots, k_n$ be nonnegative integers with sum κ. Then*

$$\sum_{\lambda \in \mathcal{C}_{k_0,\ldots,k_n}} q^{|\lambda|} = \frac{(q^{\kappa+n+1};q^{2\kappa+2n+2})_\infty (q^{2\kappa+2n+2};q^{2\kappa+2n+2})_\infty^n}{(q;q^2)_\infty (q;q)_\infty^n} \prod_{i=1}^{n} \theta(q^{\kappa_i+n-i+1};q^{\kappa+n+1}) \times \prod_{1\leq i<j\leq n} \theta(q^{\kappa_i-\kappa_j-i+j}, q^{\kappa_i+\kappa_j+2n-i-j+2};q^{2\kappa+2n+2}),$$

where $\kappa_i := k_i + \cdots + k_n$ for $1 \leq i \leq n$.

Conjecture 6.3 ([22, Conjecture 4.2]). *Let $n \geq 1$, and let $k_0, \ldots, k_n$ be nonnegative integers with sum κ. Then*

$$\sum_{\lambda \in \mathcal{D}_{k_0,\ldots,k_n}} q^{|\lambda|} = \frac{(q^{2\kappa+2n};q^{2\kappa+2n})_\infty^n}{(q^2;q^2)_\infty (q;q)_\infty^{n-1}} \prod_{1\leq i<j\leq n} \theta(q^{\kappa_i-\kappa_j-i+j}, q^{\kappa_i+\kappa_j+2n-i-j+1};q^{2\kappa+2n}),$$

where $\kappa_i := k_i + \cdots + k_n$ for $1 \leq i \leq n$.

We next consider the extremal cases in [22, Sections 3 and 4]. The level-one A-type case and the rank-one C-type case used below are established partition identities, not consequences of assuming the general conjectures. The general conjectures above fix our notation and provide context.

The established case $k_0+\cdots+k_n=1$ of Conjecture 6.1 [22, Section 3] gives, for $0\le a\le n$,

$$\sum_{\lambda\in\mathcal{A}_{0^a,1,0^{n-a}}} q^{|\lambda|}=\frac{(q^{2a+1},q^{2n-2a+2},q^{2n+3};q^{2n+3})_\infty}{(q;q)_\infty}.$$

For $n=1$, Conjecture 6.2 reduces to

$$\sum_{\lambda\in\mathcal{C}_{k_0,k_1}} q^{|\lambda|}=\frac{(q^{k_0+1},q^{k_1+1},q^{k_0+k_1+2};q^{k_0+k_1+2})_\infty}{(q;q^2)_\infty(q;q)_\infty}.$$

The product is the principal specialization obtained by Lepowsky and Milne [27, Theorem 5.9]. The corresponding partition identity follows from the coloured partition model in [33, Section 11]; see [22, Section 4] for the explicit correspondence with $\mathcal{C}_{k_0,k_1}$.

The corresponding $\kappa=1$ specialization of Conjecture 6.2 is

$$\sum_{\lambda\in\mathcal{C}_{0^a,1,0^{n-a}}} q^{|\lambda|}=\frac{(q^{2a+2},q^{2n-2a+2},q^{2n+4};q^{2n+4})_\infty}{(q;q)_\infty}.$$

This identity follows from the coloured partition theorem of Jing, Misra and Savage [19, Theorem 1.2]; the correspondence of parameters is given in [22, Section 4]. Similarly, the $\kappa=1$ specialization of Conjecture 6.3 is

$$\sum_{\lambda\in\mathcal{D}_a^{(n)}} q^{|\lambda|}=\frac{(q^{2a+1},q^{2n-2a+1},q^{2n+2};q^{2n+2})_\infty}{(q;q)_\infty},$$

where $\mathcal{D}_a^{(n)}$ denotes the case $k_a=1$ and $k_i=0$ for $i\neq a$. This identity also follows from [19, Theorem 1.2], with the parameter correspondence given in [22, Section 4].

We obtain multisum representations for two of these extremal product families by specializing Theorem 1.2.

Corollary 6.4. *Let $n\ge 1$, $0\le a\le n$, and set*

$$s_a=\begin{cases}2a, & 0\le a\le\lfloor n/2\rfloor,\\ 2n-2a+1, & \lfloor n/2\rfloor<a\le n.\end{cases}$$

Then

$$(q,q^3;q)_\infty(q^4;q^2)_\infty B_2^{s_a,n}(1,q)=\sum_{\lambda\in\mathcal{A}_{0^a,1,0^{n-a}}} q^{2|\lambda|}.$$

If k_0,k_1 are nonnegative integers, $k_0+k_1\ge 3$ is odd, and $k_0\le k_1-1$, then

$$\frac{(q,q^3;q)_\infty(q^4;q^2)_\infty}{(q^2;q^4)_\infty}B_2^{k_0,(k_0+k_1-1)/2}(1,q)=\sum_{\lambda\in\mathcal{C}_{k_0,k_1}} q^{2|\lambda|}.$$

Proof. At $z=1,w=q^2$, Theorem 1.2 gives

$$B_2^{s,k}(1,q)=\frac{\Theta_s^k}{(q,q^3;q)_\infty(q^4;q^2)_\infty(q^2;q^2)_\infty}.$$

For the first identity, take $k=n$ and $s=s_a$. The two nontrivial theta arguments are q^{4a+2} and $q^{4n-4a+4}$, possibly in reversed order. Cancelling the displayed factors gives the first product above with q replaced by q^2. For the second, take $k=(k_0+k_1-1)/2$ and $s=k_0$. The theta arguments are q^{2k_0+2}, q^{2k_1+2}, and $q^{2(k_0+k_1+2)}$. The additional divisor $(q^2;q^4)_\infty$ gives exactly the rank-one C-type denominator. □

Thus Theorem 1.2 provides multisum expressions for the two established partition generating functions. The products associated with $\mathcal{C}_{0^a,1,0^{n-a}}$ for general n, and with $\mathcal{D}_a^{(n)}$, have moduli $2n+4$ and $2n+2$, respectively, and therefore are not directly covered by Theorem 1.2. The Hall–Littlewood formulations of these partition families in [22, Conjectures 1.4 and 4.7] provide a separate point of comparison, discussed in Section 7.

7 Conclusion and further directions

We have constructed a symmetrized B_2 Bailey chain and lattice with bases q and q^2, together with two explicit Bailey pairs. Their iterations yield an A_1-type theta-product identity and a finite sum of two A_1-type theta products, respectively. The common polynomial E_s is the normalized character of $V_{B_2}(s\omega_{\mathrm{sp}})$; its finite-dimensional $D_2 = A_1 \oplus A_1$ branching rule explains the two rank-one sums in the second identity. The character specializations and CMPP product comparisons in Section 6 give further consequences.

A Hall–Littlewood reformulation. The principal specialization formula also rewrites the quadratic part of our multisums in terms of ordinary Hall–Littlewood polynomials. Let ρ, ν be partitions with $\rho_1, \nu_1 \le k$, and set

$$\lambda_i = \rho'_i, \qquad \mu_i = \nu'_i, \qquad \lambda_{k+1} = \mu_{k+1} = 0.$$

Define

$$\begin{aligned}\mathcal{H}^{B_2}_{\rho,\nu}(z,w;q) := q^{-2\sum_{i=1}^{k}\rho'_i\nu'_i}(zq)^{|\rho|}(wq^2)^{|\nu|} \\ \times P_{2\rho}(1,q,q^2,\dots;q)P_{2\nu}(1,q^2,q^4,\dots;q^2).\end{aligned} \tag{7.1}$$

The principal specialization formula [30, Chapter III], in the form used in [22, Eq. (1.10)], gives

$$(zq)^{|\rho|}P_{2\rho}(1,q,q^2,\dots;q) = \prod_{i=1}^{k}\frac{z^{\lambda_i}q^{\lambda_i^2}}{(q;q)_{\lambda_i-\lambda_{i+1}}}.$$

Applying the same formula with base q^2 gives

$$\mathcal{H}^{B_2}_{\rho,\nu}(z,w;q) = \prod_{i=1}^{k}\frac{z^{\lambda_i}w^{\mu_i}q^{\lambda_i^2-2\lambda_i\mu_i+2\mu_i^2}}{(q;q)_{\lambda_i-\lambda_{i+1}}(q^2;q^2)_{\mu_i-\mu_{i+1}}}. \tag{7.2}$$

The summands in Theorems 1.2 and 1.4 are obtained by multiplying (7.2) by the appropriate terminal factor

$$\widehat{\mathcal{A}}_{N,L} := (q;q)_N(q^2;q^2)_L\mathcal{A}_{N,L}, \qquad \widehat{\mathcal{R}}_{N,L} := (q;q)_N(q^2;q^2)_L\mathcal{R}_{N,L},$$

evaluated at $N = \rho'_k$ and $L = \nu'_k$, and by the linear or three-term factor already present in the theorem.

Further questions. It remains to find additional B_2 Bailey pairs, lattice transformations that produce other moduli, including cases of the CMPP products not covered by Theorem 1.2. Another question is whether the coupled expression $\mathcal{H}^{B_2}_{\rho,\nu}$ can be obtained from a BC_2- or C_2-type Hall–Littlewood identity. Such an interpretation would need to account for the interaction factor and the terminal factors in addition to the two ordinary principal specializations in (7.1).

A A projection of Warnaar's A_2 Bailey tree

The following identity is independent of the B_2 construction. It follows from a one-directional specialization of Warnaar's A_2 Bailey tree [40].

Theorem A.1. *Let $k \geq 1$ and $0 \leq a \leq k$ be integers. For generic nonzero z, w, with $0 < |q| < 1$ and all denominators nonzero, we have*

$$\sum_{\substack{\lambda_1 \geq \cdots \geq \lambda_k \in \mathbb{Z} \\ \mu_1 \geq \cdots \geq \mu_k \geq 0}} \frac{z^{\sum_{i=1}^k \lambda_i} w^{\sum_{i=1}^k \mu_i} q^{\sum_{i=1}^k (\lambda_i^2 - \lambda_i \mu_i + \mu_i^2) - \sum_{i=1}^a \lambda_i}}{\prod_{i=1}^{k-1} (q;q)_{\lambda_i - \lambda_{i+1}} (q;q)_{\mu_i - \mu_{i+1}}} \frac{1}{(zwq;q)_{\lambda_k} (q;q)_{\mu_k}}$$

$$= \frac{(q^{2k+4};q^{2k+4})_\infty}{(1-z)(zq, q/z, zwq;q)_\infty} \sum_{r=0}^{a} \sum_{s=0}^{r} z^r w^s \Big[\theta(-z^{k+2} q^{k+1+2r-s-a}; q^{2k+4}) - z\theta(-z^{k+2} q^{k+3+2r-s-a}; q^{2k+4}) \Big]. \tag{A.1}$$

We use Warnaar's A_2 kernel and rational function [40, 41]:

$$K_{n,m;r,\ell}(z,w;q) = \frac{z^r w^\ell q^{r^2 - r\ell + \ell^2}}{(q;q)_{n-r} (q;q)_{m-\ell}},$$
$$\Phi_{n,m}(z,w;q) = \frac{(zwq;q)_{n+m}}{(q, zq, zwq;q)_n (q, wq, zwq;q)_m}. \tag{A.2}$$

The two-parameter deformation from [40, Eq. (4.10)] is

$$\Phi_{n,m}(U,V;z,w;q) = \Phi_{n,m}(z/q, w;q) - \frac{Uz}{(z;q)_2} \Phi_{n-1,m}(zq, w/q;q) + \frac{UVzw^2}{(w, zw;q)_2} \Phi_{n-1,m-1}(z, wq;q). \tag{A.3}$$

In particular, $\Phi_{n,m}(1,1;z,w;q) = \Phi_{n,m}(z,w;q)$. For $y \in \mathbb{Z}$, put

$$\Phi^*_{n,m;y}(U,V;z,w;q) := \frac{\Phi_{n-y,m}(U,V; zq^{2y}, wq^{-y};q)}{(zq;q)_{2y} (wq;q)_{-y} (zwq;q)_y}. \tag{A.4}$$

Only the following specialization of Warnaar's shifted Bailey tree is needed. In the notation of [40, Corollary 4.6], take the root-lattice vector $(y, -y, 0)$.

Theorem A.2 ([40, Corollary 4.6], specialized)**.** *For integers n, m, y, we have*

$$\sum_{r=y}^{n} \sum_{\ell=0}^{m} K_{n,m;r,\ell}(z,w;q) \Phi^*_{r,\ell;y}(1,1;z,w;q) = z^y q^{y^2} \Phi^*_{n,m;y}(1,1;z,w;q), \tag{A.5a}$$

$$\sum_{r=y}^{n} \sum_{\ell=0}^{m} K_{n,m;r,\ell}(z/q,w;q) \Phi^*_{r,\ell;y}(U,V;z,w;q) = z^y q^{y^2-y} \Phi^*_{n,m;y}(Uzq^{2y}, Vwq^{-y};z,w;q). \tag{A.5b}$$

Proof of Theorem A.1. Apply (A.5a) $k-a$ times and then (A.5b) a times. Writing $\lambda_0 = n$ and $\mu_0 = m$, and using $K_{n,m;r,\ell}(z/q,w;q) = q^{-r} K_{n,m;r,\ell}(z,w;q)$, gives

$$\sum_{\substack{n=\lambda_0 \geq \cdots \geq \lambda_k \geq y \\ m=\mu_0 \geq \cdots \geq \mu_k \geq 0}} q^{-\sum_{i=1}^a \lambda_i} \prod_{i=1}^{k} K_{\lambda_{i-1},\mu_{i-1};\lambda_i,\mu_i}(z,w;q) \Phi^*_{\lambda_k,\mu_k;y}(1,1;z,w;q)$$

$$= z^{ky} q^{ky^2-ay} \Phi^*_{n,m;y}\big((zq^{2y})^a, (wq^{-y})^a; z, w; q\big). \tag{A.6}$$

For later use, the undeformed shifted function can be written as

$$\Phi^*_{r,s;y}(1,1;z,w;q) = \frac{(zwq;q)_{r+s}}{(q;q)_{r-y}(zq;q)_{r+y}(zwq;q)_r(q;q)_s(wq;q)_{s-y}(zwq;q)_{s+y}}.$$

This follows by cancelling the shifted factorials in (A.4); it makes the subsequent bilateral summation explicit.

Set $A_y(z) = (1-zq^{2y})z^{2y}q^{2y^2-y}$. The specialization of Bailey's ${}_6\psi_6$ summation (2.3) used here is

$$\sum_{y\in\mathbb{Z}} \frac{A_y(z)}{(q;q)_{r-y}(wq;q)_{s-y}(zq;q)_{r+y}(zwq;q)_{s+y}} = \frac{(1-z)(q/z;q)_\infty}{(wq;q)_\infty(zwq;q)_{r+s}}. \tag{A.7}$$

Here (A.7) follows by taking $a=z$, $b=q^{-r}$, $c=q^{-s}/w$ and letting the remaining two numerator parameters tend to infinity in (2.3).

Multiply (A.6) by $A_y(z)$ and sum over $y\in\mathbb{Z}$. By (A.7), the limit of the left-hand side as $n,m\to\infty$ is the multisum in (A.1) multiplied by $(1-z)(q/z;q)_\infty/((wq;q)_\infty(q;q)_\infty^2)$.

For the right-hand side, write $Z=zq^{2y}$ and $W=wq^{-y}$. From (A.3),

$$\lim_{n,m\to\infty} \Phi_{n,m}(U,V;Z,W;q) = \frac{(1-W)-UZ(1-ZW)+UVZW^2(1-Z)}{(q;q)_\infty^2(Z,W,ZW;q)_\infty}. \tag{A.8}$$

Combining this with (A.4) and simplifying gives

$$\begin{aligned} &\lim_{n,m\to\infty} \Phi^*_{n,m;y}(Z^a,W^a;z,w;q) = \frac{S_a(Z,W)}{D(z,w;q)}, \quad D(z,w;q) = (q;q)_\infty^2(zq,wq,zwq;q)_\infty, \\ &S_a(Z,W) = \frac{(1-W)-Z^{a+1}(1-ZW)+Z^{a+1}W^{a+2}(1-Z)}{(1-Z)(1-W)(1-ZW)} = \sum_{r=0}^{a}\sum_{s=0}^{r} Z^rW^s. \end{aligned} \tag{A.9}$$

Thus the limit of the y-summed right-hand side of (A.6) is

$$\frac{1}{D(z,w;q)} \sum_{r=0}^{a}\sum_{s=0}^{r} z^r w^s \sum_{y\in\mathbb{Z}} z^{(k+2)y} q^{(k+2)y^2+(2r-s-a-1)y}(1-zq^{2y}). \tag{A.10}$$

Finally, the Jacobi triple product gives

$$\sum_{y\in\mathbb{Z}} z^{(k+2)y} q^{(k+2)y^2+by} = (q^{2k+4};q^{2k+4})_\infty \theta(-z^{k+2}q^{k+2+b};q^{2k+4}). \tag{A.11}$$

Apply this identity with $b=2r-s-a-1$ and $b=2r-s-a+1$ to (A.10). Equating the result with the limit of the left-hand side described above, and substituting $D(z,w;q)$, gives (A.1). □

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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